

The Modern AI Concept Book

Edition 2 · September 2026

LLMs, VLMs, agents, harnesses, hardware and the practice of AI engineering.

Current to September 20, 2026.

17 categories 173 concepts 430 dated resources

Resource types: ■ GitHub ■ Blog ■ Paper ■ Docs ■ Spec ■ Video

How to use this book

Each entry gives a plain-language definition, key points, a *state of the art* note where the picture changed in 2026, and dated resources (GitHub repositories, blogs, papers, documentation, specifications and videos).

The header line under each concept shows the year it emerged, its most recent active year, its level (Beginner, Intermediate, Advanced) and tags. **NEW 2026** marks concepts that appeared in 2026; **ACTIVE 2026** marks older concepts that were still changing in 2026.

A printed book cannot be filtered, so the appendices provide the same cuts as the interactive HTML edition: concepts by year of emergence, concepts by level, all resources grouped by type and sorted by year, and an index of acronyms. For live filtering by category, year, level, tag, resource type and resource year, open the HTML edition or the spreadsheet catalog.

Resource years show when an article, paper or specification was published, or when a repository or site launched (approximate for long-running projects). Chapter 1 reflects public reporting as of September 20, 2026 and will date quickly.

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1

The Frontier Landscape (September 2026)

A snapshot of where frontier and open-weight models stand as of September 2026. This chapter dates quickly; use the trackers at the end to stay current.

Mythos-class tier and trusted access (Fable / Mythos)

NEW 2026 Since 2026 Beginner Enterprise, Security

Anthropic introduced a capability tier above Opus. The same underlying model ships in two forms: Claude Fable for general use with extra safeguards in biology, cybersecurity and LLM R&D, and Claude Mythos for vetted defenders and partners through Project Glasswing.

- Mythos Preview (April 2026) was first restricted to trusted organisations for cyber defense.
- Fable 5 / Mythos 5 launched June 9, 2026; access was suspended June 12 to comply with US export controls and restored July 1 after the controls were lifted.
- Fable 5.1 and Mythos 5.1 followed on September 1, 2026, with 1M-token context.

STATE OF THE ART, SEPT 2026

Fable 5.1 is Anthropic's generally available flagship; Opus 5 (July 2026) and Sonnet 5 sit below it.

■ **Docs** 2026 Anthropic: Project Glasswing

<https://www.anthropic.com/glasswing>

■ **Docs** 2026 Anthropic statement on Fable and Mythos access

<https://www.anthropic.com/news/fable-mythos-access>

■ **Blog** 2026 DataNorth: Claude Fable 5 and Mythos 5 explained

<https://datanorth.ai/news/anthropic-releases-claude-fable-5-and-mythos-5>

GPT-6 Astra and the Critical cyber threshold (GPT-6)

NEW 2026 Since 2026 Beginner Security, Enterprise

OpenAI's first GPT-6 generation model, released September 3, 2026. It is the first model OpenAI classified at the Critical cybersecurity level of its Preparedness Framework, so it launched in phases with stronger safeguards and a trusted-access program for defenders.

- Critical means the model can find and exploit unknown vulnerabilities in hardened systems without step-by-step human guidance.
- Public versions refuse advanced offensive tasks; vetted defenders get more through OpenAI's trusted-access programs.
- Its system card also reported reduced chain-of-thought monitorability, a notable safety signal.

STATE OF THE ART, SEPT 2026

GPT-6 Astra sits above the GPT-5.6 Sol / Terra / Luna tiers.

■ **Blog** 2026 OpenAI: Path to Astra

<https://openai.com/index/path-to-astra/>

■ **Docs** 2026 GPT-6 Astra system card

<https://deploymentsafety.openai.com/gpt-6-astra>

■ **Blog** 2026 OpenAI: Safety overview of GPT-6 Astra

<https://openai.com/index/safety-overview-gpt-6-astra/>

■ **Blog** 2026 InfoQ: Astra is first model rated Critical for cyber

<https://www.infoq.com/news/2026/09/gpt-6-astra-critical-cyber/>

Tiered access for dual-use models (trusted access)

ACTIVE 2026 Since 2025–2026 Intermediate Security, Enterprise

A 2026 industry pattern: the most cyber-capable models are released in a public version with guardrails and a restricted version for vetted security teams and critical-infrastructure operators.

- Anthropic (Glasswing), OpenAI (trusted access for cyber) and Google (defender-only cyber variants) all adopted versions of it.
- It lets defenders patch vulnerabilities before equivalent capabilities spread.
- Enterprises now need to track which model variant, safeguards and data-retention terms they are using.

■ **Docs** 2026 [Anthropic: Project Glasswing](https://www.anthropic.com/glasswing)
<https://www.anthropic.com/glasswing>

■ **Blog** 2026 [CSO Online: Astra crosses the Critical cyber threshold](https://www.csoonline.com/article/4218679/openai-launches-gpt-6-astra-its-first-model-to-cross-a-critical-cybersecurity-threshold.html)
<https://www.csoonline.com/article/4218679/openai-launches-gpt-6-astra-its-first-model-to-cross-a-critical-cybersecurity-threshold.html>

Open-weight frontier models

ACTIVE 2026 Since 2023–2026 Beginner Open source / open weights, Cost & efficiency

Open-weight models trail the closed frontier by months, not years. 2026's leading families are mostly large sparse MoEs with 1M-token context and permissive licences.

- DeepSeek V4 (April 2026): Pro has 1.6T total / 49B active parameters, Flash has 284B / 13B, both with 1M context, MIT licence.
- Other active families: Qwen 3.x, Kimi, GLM-5.x, MiniMax, Mistral Large 3, NVIDIA Nemotron 3, Google Gemma 4.
- Open weight does not always mean open source: check licence terms and whether data and code are released.

STATE OF THE ART, SEPT 2026

DeepSeek V4 is the reference open model for long-context efficiency; small open models now run strong coding workloads on a single GPU.

■ **Paper** 2026 [DeepSeek-V4 technical report](https://arxiv.org/abs/2606.19348)
<https://arxiv.org/abs/2606.19348>

■ **Blog** 2026 [HF blog: DeepSeek V4 architecture analysis](https://huggingface.co/blog/ResterChed/deepseek-v4-ga-architecture)
<https://huggingface.co/blog/ResterChed/deepseek-v4-ga-architecture>

■ **Docs** 2016 [Hugging Face models hub](https://huggingface.co/models)
<https://huggingface.co/models>

Model release trackers

ACTIVE 2026 Since 2023–2026 Beginner Production practice

In 2026 frontier labs shipped new models every few weeks. Trackers and independent benchmark aggregators are the practical way to keep up.

- Compare on your own evals, not headline benchmarks.
- Watch pricing changes as closely as capability changes: cache-read and small-tier prices fell sharply in 2026.

■ **Docs** 2024 [Artificial Analysis](https://artificialanalysis.ai)
<https://artificialanalysis.ai>

■ **Docs** 2024 [Epoch AI benchmarking hub](https://epoch.ai/benchmarks)
<https://epoch.ai/benchmarks>

■ **Docs** 2023 [LMArena](https://lmarena.ai)
<https://lmarena.ai>

■ **Docs** 2023 [OpenRouter model list](https://openrouter.ai/models)
<https://openrouter.ai/models>

2

Agents and Harnesses

An agent is a model running in a loop: it reads context, chooses an action, calls a tool, observes the result and repeats. The industry now summarises this as Agent = Model + Harness, and most 2026 engineering effort goes into the harness.

Agent harness

ACTIVE 2026 Since 2024–2026 Beginner Foundational, Production practice

Everything around the model that turns it into a working agent: the loop, tool definitions, permissions, sandbox, context management, memory, retries, logging and human approval gates.

- Two teams using the same model can get very different results purely from harness design.
- Harness setup alone can move benchmark scores by several points.
- Harnesses encode assumptions about model weaknesses, so they must be revisited when models improve.

STATE OF THE ART, SEPT 2026

Vendor harnesses (Claude Code, Codex, Gemini CLI) are now exposed as SDKs and managed runtimes so teams can build on them directly.

■ **Blog** 2026 Martin Fowler: Harness engineering for coding agent users

<https://martinfowler.com/articles/harness-engineering.html>

■ **Blog** 2025 Anthropic: Effective harnesses for long-running agents

<https://www.anthropic.com/engineering/effective-harnesses-for-long-running-agents>

■ **GitHub** 2026 awesome-harness-engineering

<https://github.com/ai-boost/awesome-harness-engineering>

Harness engineering

NEW 2026 Since 2026 Intermediate Production practice

The discipline that followed prompt engineering and context engineering: designing environments, constraints, documentation-as-system-of-record, structural tests and feedback loops so agents do reliable work while humans steer.

- OpenAI described a team shipping a roughly million-line product with no hand-written code using this approach.
- Core moves: constrain what agents can do, inform them what to do, verify their work, correct mistakes, keep humans at high-stakes gates.
- Architecture rules become linters and tests the agent must pass, not wiki pages.

STATE OF THE ART, SEPT 2026

The term entered mainstream use in early 2026 and is now the main focus of agent engineering investment.

■ **Blog** 2026 OpenAI: Harness engineering

<https://openai.com/index/harness-engineering/>

■ **Blog** 2026 Faros: Harness engineering guide

<https://www.faros.ai/blog/harness-engineering>

■ **GitHub** 2026 awesome-agent-harness

<https://github.com/AutoJunjie/awesome-agent-harness>

Managed agents: decoupling brain, hands and session

NEW 2026 Since 2026 Advanced Production practice, Enterprise

An architecture where the reasoning loop (brain), the sandboxes and tools that act (hands), and the append-only event log (session) are separate, replaceable services.

- The harness becomes stateless; containers are provisioned only when a tool needs them.
- A crash in one sandbox does not lose the session, because state lives in the log.
- Enterprises can run the hands inside their own network while the brain runs at the provider.

STATE OF THE ART, SEPT 2026

Anthropic launched Claude Managed Agents on this design in April 2026; Cursor's cloud agents converged on a similar three-layer split.

■ **Blog** 2026 [Anthropic: Scaling Managed Agents](https://anthropic.com/engineering/managed-agents)
<https://anthropic.com/engineering/managed-agents>

Agentic loop / ReAct (ReAct)

Since 2022–2025 Beginner Foundational

The basic control pattern: reason about the next step, act by calling a tool, observe the result, and loop until done.

- Modern models interleave thinking between tool calls natively.
- A usable coding agent loop fits in about 100 lines of code.

■ **Paper** 2022 [ReAct paper](https://arxiv.org/abs/2210.03629)
<https://arxiv.org/abs/2210.03629>

■ **Blog** 2024 [Anthropic: Building effective agents](https://www.anthropic.com/engineering/building-effective-agents)
<https://www.anthropic.com/engineering/building-effective-agents>

■ **GitHub** 2025 [mini-swe-agent](https://github.com/SWE-agent/mini-swe-agent)
<https://github.com/SWE-agent/mini-swe-agent>

Workflows vs. agents

Since 2024–2025 Beginner Foundational, Production practice

Workflows run LLM calls through predefined code paths; agents let the model direct its own process. Use the simplest pattern that works.

- Common workflow patterns: prompt chaining, routing, parallelisation, orchestrator-workers, evaluator-optimiser.
- Add autonomy only when the path cannot be predicted in advance.

■ **Blog** 2024 [Anthropic: Building effective agents](https://www.anthropic.com/engineering/building-effective-agents)
<https://www.anthropic.com/engineering/building-effective-agents>

■ **GitHub** 2025 [12-factor agents](https://github.com/humanlayer/12-factor-agents)
<https://github.com/humanlayer/12-factor-agents>

Tool use / function calling

Since 2023–2025 Beginner Foundational

Models emit structured calls (a tool name plus JSON arguments) that the harness executes and returns. Tool design is API design for a non-human user.

- Prefer few, expressive tools over many narrow ones.
- Return compact, high-signal output and actionable error messages.
- Namespacing and clear descriptions reduce wrong-tool calls.

■ **Blog** 2025 [Anthropic: Writing effective tools for agents](https://www.anthropic.com/engineering/writing-tools-for-agents)
<https://www.anthropic.com/engineering/writing-tools-for-agents>

■ **Docs** 2024 [Claude tool use docs](https://docs.claude.com/en/docs/agents-and-tools/tool-use/overview)
<https://docs.claude.com/en/docs/agents-and-tools/tool-use/overview>

Tool search and programmatic tool calling (code mode)

ACTIVE 2026 Since 2025–2026 Intermediate Cost & efficiency, Production practice

Two fixes for tool overload. Tool search loads tool definitions on demand; programmatic tool calling lets the model write a script that calls many tools and returns only the final result.

- Cuts context use and round-trips dramatically for large tool sets.
- Deferred tool discovery and stable tool ordering also protect prompt-cache hits.

STATE OF THE ART, SEPT 2026

Now a standard feature in frontier harnesses; MCP servers are increasingly consumed through code rather than direct calls.

- **Blog** 2025 [Anthropic: Advanced tool use](https://www.anthropic.com/engineering/advanced-tool-use)
<https://www.anthropic.com/engineering/advanced-tool-use>
- **Blog** 2025 [Anthropic: Code execution with MCP](https://www.anthropic.com/engineering/code-execution-with-mcp)
<https://www.anthropic.com/engineering/code-execution-with-mcp>
- **Blog** 2025 [Cloudflare: Code Mode](https://blog.cloudflare.com/code-mode/)
<https://blog.cloudflare.com/code-mode/>

Multi-agent systems and subagents (orchestrator-worker)

ACTIVE 2026 Since 2023–2026 Intermediate Production practice

A lead agent plans and delegates to subagents with their own clean context windows, then combines condensed results.

- Great for broad research and big codebases; costs more tokens.
- Specialised orchestrator models can route work between tools, small models and large models.

STATE OF THE ART, SEPT 2026

Agent teams, swarms and parallel agents in isolated worktrees are routine in 2026 coding tools.

- **Blog** 2025 [Anthropic: Multi-agent research system](https://www.anthropic.com/engineering/multi-agent-research-system)
<https://www.anthropic.com/engineering/multi-agent-research-system>
- **Docs** 2025 [Claude Code subagents](https://docs.claude.com/en/docs/claude-code/sub-agents)
<https://docs.claude.com/en/docs/claude-code/sub-agents>
- **GitHub** 2024 [LangGraph](https://github.com/langchain-ai/langgraph)
<https://github.com/langchain-ai/langgraph>

Long-running and background agents

ACTIVE 2026 Since 2025–2026 Intermediate Production practice

Agents that work for hours or days across many context windows, usually in the cloud while the human does something else.

- Rely on plan files, progress notes, git history, test gates and compaction.
- Early models showed context anxiety (wrapping up early near the limit); newer models largely don't, so harness workarounds can become dead weight.

STATE OF THE ART, SEPT 2026

METR's time-horizon measure keeps doubling; multi-hour autonomous runs are normal for frontier models.

- **Blog** 2025 [Anthropic: Effective harnesses for long-running agents](https://www.anthropic.com/engineering/effective-harnesses-for-long-running-agents)
<https://www.anthropic.com/engineering/effective-harnesses-for-long-running-agents>
- **Blog** 2025 [METR: Measuring AI ability to complete long tasks](https://metr.org/blog/2025-03-19-measuring-ai-ability-to-complete-long-tasks/)
<https://metr.org/blog/2025-03-19-measuring-ai-ability-to-complete-long-tasks/>

Computer use and browser agents (CUA)

ACTIVE 2026 Since 2024–2026 Intermediate Production practice

Agents that operate a GUI from screenshots with mouse and keyboard actions, or drive a browser through the DOM and accessibility tree.

- Agentic browsers embed this directly in the browser.
- Prompt injection from web content is the main security risk.

■ **Docs** 2024 [Claude computer use tool](#)

<https://docs.claude.com/en/docs/agents-and-tools/tool-use/computer-use-tool>

■ **Blog** 2025 [OpenAI: Computer-Using Agent](#)

<https://openai.com/index/computer-using-agent/>

■ **GitHub** 2024 [browser-use](#)

<https://github.com/browser-use/browser-use>

■ **GitHub** 2025 [Playwright MCP](#)

<https://github.com/microsoft/playwright-mcp>

Personal agents (OpenClaw)

NEW 2026 Since 2026 Beginner Open source / open weights, Security

Always-on assistants connected to messaging apps, email, calendars and files that take real actions for a person.

- OpenClaw’s viral open-source spread in early 2026 put millions of personal agents into daily use.
- Exposed the permission, isolation and injection problems of agents with broad personal access.

■ **GitHub** 2026 [OpenClaw](#)

<https://github.com/openclaw/openclaw>

■ **Paper** 2026 [SemaClaw: personal agents via harness engineering](#)

<https://arxiv.org/abs/2604.11548>

Agent sandboxes

ACTIVE 2026 Since 2024–2026 Intermediate Security, Production practice

Isolated environments (containers, microVMs, V8 isolates, OS-level sandboxes) where agents run code without endangering the host.

- Filesystem boundaries and network egress controls are the two key levers.
- Good sandboxing reduces approval fatigue: fewer prompts, more safety.

STATE OF THE ART, SEPT 2026

Providers now let enterprises run agent sandboxes in their own clouds while the model runs remotely.

■ **GitHub** 2025 [Anthropic sandbox-runtime](#)

<https://github.com/anthropic-experimental/sandbox-runtime>

■ **GitHub** 2023 [E2B](#)

<https://github.com/e2b-dev/E2B>

■ **GitHub** 2024 [Daytona](#)

<https://github.com/daytonaio/daytona>

■ **GitHub** 2025 [container-use](#)

<https://github.com/dagger/container-use>

Agent frameworks and SDKs

ACTIVE 2026 Since 2023–2026 Beginner Open source / open weights

Libraries that package the loop, tools, memory, handoffs and tracing.

- Vendor SDKs expose the same harness their own products use.
- Open frameworks add graphs, durable execution and multi-agent patterns.

■ **GitHub** 2025 [Claude Agent SDK \(Python\)](#)

<https://github.com/anthropics/claude-agent-sdk-python>

- **GitHub** 2025 [OpenAI Agents SDK](https://github.com/openai/openai-agents-python)
<https://github.com/openai/openai-agents-python>
- **GitHub** 2025 [Google ADK](https://github.com/google/adk-python)
<https://github.com/google/adk-python>
- **GitHub** 2025 [Microsoft Agent Framework](https://github.com/microsoft/agent-framework)
<https://github.com/microsoft/agent-framework>
- **GitHub** 2024 [Pydantic AI](https://github.com/pydantic/pydantic-ai)
<https://github.com/pydantic/pydantic-ai>
- **GitHub** 2024 [smolagents](https://github.com/huggingface/smolagents)
<https://github.com/huggingface/smolagents>
- **GitHub** 2023 [CrewAI](https://github.com/crewAIInc/crewAI)
<https://github.com/crewAIInc/crewAI>
- **GitHub** 2025 [LangChain deepagents](https://github.com/langchain-ai/deepagents)
<https://github.com/langchain-ai/deepagents>

Durable execution and agent trajectories

ACTIVE 2026 Since 2024–2026 Advanced Production practice

Persist every step so an agent can crash, pause for approval or wait days, then resume. Trajectory logs also feed memory, evals and training.

- Workflow engines like Temporal are used as agent backbones.
- Libraries now normalise session transcripts from many harnesses into one format.

- **Docs** 2019 [Temporal](https://temporal.io)
<https://temporal.io>
- **GitHub** 2024 [LangGraph \(persistence\)](https://github.com/langchain-ai/langgraph)
<https://github.com/langchain-ai/langgraph>

3

Protocols and Interoperability Standards

Open protocols let agents, tools, apps, editors and payment systems interoperate without bespoke glue. Several now live in neutral foundations.

Model Context Protocol (MCP)

ACTIVE 2026 Since 2024–2026 Beginner Open standard, Foundational

An open JSON-RPC protocol connecting AI applications (clients) to tools, resources and prompts exposed by servers. The de facto standard for tool integration across vendors.

- 2026-07-28 revision: stateless core (no sessions or initialize handshake), server discovery on demand, cacheable list results, header-based routing.
- Formal Extensions framework; MCP Apps and Tasks ship as extensions.
- Authorization aligned with OAuth/OIDC: issuer validation, Client ID Metadata Documents, enterprise-managed authorization.

STATE OF THE ART, SEPT 2026

The 2026-07-28 spec is the largest revision since launch, with a 12-month deprecation policy and SDK conformance tiers.

- **Blog** 2026 [MCP blog: The 2026-07-28 specification](https://blog.modelcontextprotocol.io/posts/2026-07-28/)
<https://blog.modelcontextprotocol.io/posts/2026-07-28/>
- **Blog** 2026 [MCP roadmap \(Aug 2026\)](https://blog.modelcontextprotocol.io/posts/mcp-roadmap/)
<https://blog.modelcontextprotocol.io/posts/mcp-roadmap/>
- **Docs** 2024 modelcontextprotocol.io
<https://modelcontextprotocol.io>
- **GitHub** 2024 [MCP org \(SDKs, servers\)](https://github.com/modelcontextprotocol)
<https://github.com/modelcontextprotocol>
- **Blog** 2026 [WorkOS: What changes for agent authentication](https://workos.com/blog/mcp-2026-spec-agent-authentication)
<https://workos.com/blog/mcp-2026-spec-agent-authentication>

MCP Apps

ACTIVE 2026 Since 2025–2026 Intermediate Open standard

An MCP extension that lets servers return interactive UI rendered in a sandboxed frame, so tools can show forms, charts and pickers inside the chat.

- Standardised as an extension in the 2026-07-28 release.

- **GitHub** 2025 [MCP ext-apps](https://github.com/modelcontextprotocol/ext-apps)
<https://github.com/modelcontextprotocol/ext-apps>

MCP Tasks

ACTIVE 2026 Since 2025–2026 Intermediate Open standard

An extension for long-running work: a server can accept a job, stream progress and let the client check back or steer mid-flight.

- Paired with subscriptions and progress notifications for push-style updates.

- **Blog** 2026 [MCP release candidate notes](https://blog.modelcontextprotocol.io/posts/2026-07-28-release-candidate/)
<https://blog.modelcontextprotocol.io/posts/2026-07-28-release-candidate/>

Agent2Agent protocol (A2A)

ACTIVE 2026 Since 2025–2026 Intermediate Open standard

An open protocol under the Linux Foundation for agents from different vendors to discover each other via Agent Cards, exchange tasks and stream results.

- MCP is agent-to-tool; A2A is agent-to-agent.

■ **Spec** 2025 [A2A protocol](https://a2a-protocol.org)
<https://a2a-protocol.org>

■ **GitHub** 2025 [A2A repository](https://github.com/a2aproject/A2A)
<https://github.com/a2aproject/A2A>

Agentic AI Foundation (AAIF)

ACTIVE 2026 Since 2025–2026 Beginner Open standard

A Linux Foundation home for vendor-neutral agent standards, founded December 2025 with MCP, AGENTS.md and goose.

- Neutral governance so no single vendor controls core agent standards.
- Hosts specs, reference implementations and working groups.

■ **Docs** 2025 aaif.io
<https://aaif.io>

■ **GitHub** 2025 [goose](https://github.com/block/goose)
<https://github.com/block/goose>

Agent Skills (SKILL.md)

ACTIVE 2026 Since 2025–2026 Beginner Open standard, Production practice

Folders with a SKILL.md (name, description, instructions) plus optional scripts that an agent loads only when relevant (progressive disclosure).

- Published as an open standard; the same skill works across agent products.
- Cheap way to package team know-how, house style and workflows.

■ **Spec** 2025 agentskills.io
<https://agentskills.io>

■ **GitHub** 2025 [anthropics/skills](https://github.com/anthropics/skills)
<https://github.com/anthropics/skills>

■ **Blog** 2025 [Anthropic: Equipping agents with Agent Skills](https://www.anthropic.com/engineering/equipping-agents-for-the-real-world-with-agent-skills)
<https://www.anthropic.com/engineering/equipping-agents-for-the-real-world-with-agent-skills>

AGENTS.md

ACTIVE 2026 Since 2025–2026 Beginner Open standard

A Markdown file at the repo root telling coding agents how to build, test and follow conventions: a README for agents.

- CLAUDE.md plays the same role in Claude Code.

■ **Spec** 2025 agents.md
<https://agents.md>

■ **GitHub** 2025 [agentsmd/agents.md](https://github.com/agentsmd/agents.md)
<https://github.com/agentsmd/agents.md>

Agent Client Protocol (ACP)

ACTIVE 2026 Since 2025–2026 Intermediate Open standard

An editor-to-agent protocol similar in spirit to LSP, so any coding agent can plug into any editor.

- Editors act as clients; agents run as subprocesses speaking JSON-RPC.
- Lets one agent work in Zed, JetBrains, Neovim and others without per-editor plugins.

■ **Spec** 2025 agentclientprotocol.com

<https://agentclientprotocol.com>

■ **GitHub** 2025 [agent-client-protocol](https://github.com/zed-industries/agent-client-protocol)

<https://github.com/zed-industries/agent-client-protocol>

AG-UI

Since 2025 Intermediate Open standard

An event-based protocol for streaming agent state, tool calls and UI updates between backends and frontends.

- Streams typed events: text deltas, tool calls, state patches, human-input requests.
- Complements MCP (tools) and A2A (agents) on the user-facing side.

■ **Docs** 2025 [AG-UI docs](https://docs.ag-ui.com)

<https://docs.ag-ui.com>

■ **GitHub** 2025 [ag-ui](https://github.com/ag-ui-protocol/ag-ui)

<https://github.com/ag-ui-protocol/ag-ui>

Agentic commerce protocols (ACP, AP2, UCP, x402)

ACTIVE 2026 Since 2025–2026 Intermediate Open standard, Enterprise

Standards that let agents shop and pay for users: checkout (Agentic Commerce Protocol), verifiable payment mandates (AP2), merchant integration (UCP) and machine-to-machine payments over HTTP 402 (x402).

- Mandates prove the user authorised a purchase.
- x402 enables per-request payments between agents and APIs.

■ **GitHub** 2025 [Agentic Commerce Protocol](https://github.com/agentic-commerce-protocol/agentic-commerce-protocol)

<https://github.com/agentic-commerce-protocol/agentic-commerce-protocol>

■ **GitHub** 2025 [AP2](https://github.com/google-agentic-commerce/AP2)

<https://github.com/google-agentic-commerce/AP2>

■ **Spec** 2026 [Universal Commerce Protocol](https://ucp.dev)

<https://ucp.dev>

■ **GitHub** 2025 [x402](https://github.com/coinbase/x402)

<https://github.com/coinbase/x402>

llms.txt

Since 2024–2025 Beginner Open standard

A proposed file at a website root that gives LLMs a curated Markdown map of the site's most useful content.

- Plain Markdown with links and short descriptions.
- Often paired with .md versions of docs pages for clean agent reading.

■ **Spec** 2024 llmstxt.org

<https://llmstxt.org>

OpenTelemetry GenAI conventions (OTel GenAI)

ACTIVE 2026 Since 2024–2026 Intermediate Open standard, Production practice

Standard span and attribute names for LLM calls, tokens, tool invocations and agent steps, so traces from any framework land in any backend.

- Standard attributes for model name, token counts, prompts and tool calls.
- Supported by most LLM Ops tools and agent SDKs.

■ **Spec** 2024 OTel GenAI semantic conventions

<https://opentelemetry.io/docs/specs/semconv/gen-ai/>

4

Agentic Coding and Developer Workflow

Software engineering was the first domain where agents became daily tools at scale. This vocabulary describes how teams now work with them.

Coding agents

ACTIVE 2026 Since 2024–2026 Beginner Production practice, Open source / open weights

Terminal, IDE, desktop and cloud agents that read a repo, edit files, run commands and tests, and open pull requests.

- They differ mostly in harness, permissions and UX, not the loop itself.
- Cross-vendor plugins appeared: e.g. a Codex plugin for Claude Code.

STATE OF THE ART, SEPT 2026

Claude Code, Codex, Gemini CLI, Cursor, Google Antigravity and open tools like OpenHands and OpenCode are the main options in 2026.

- **GitHub** 2025 **Claude Code**
<https://github.com/anthropics/claude-code>
- **GitHub** 2025 **OpenAI Codex CLI**
<https://github.com/openai/codex>
- **GitHub** 2025 **Gemini CLI**
<https://github.com/google-gemini/gemini-cli>
- **GitHub** 2024 **OpenHands**
<https://github.com/OpenHands/OpenHands>
- **GitHub** 2025 **OpenCode**
<https://github.com/sst/opencode>
- **GitHub** 2023 **Aider**
<https://github.com/Aider-AI/aider>

Vibe coding

Since 2025 Beginner

Coined by Andrej Karpathy: building software by describing it and accepting AI output without reading it closely. Fine for prototypes, risky for production.

- Useful for throwaway prototypes, personal tools and learning.
- Unreviewed code carries security and maintenance risk in production.

- **Blog** 2025 **Karpathy's original post**
<https://x.com/karpathy/status/1886192184808149383>
- **Blog** 2025 **Simon Willison: Not all AI-assisted programming is vibe coding**
<https://simonwillison.net/2025/Mar/19/vibe-coding/>

Agentic engineering / Software 3.0

ACTIVE 2026 Since 2025–2026 Intermediate Production practice

The professional counterpart to vibe coding: engineers specify intent, design constraints and review outcomes while agents write most of the code.

- Humans steer, agents execute.
- Prompts in English become a new kind of program (Software 3.0).

■ **Video** 2025 [Karpathy: Software is changing \(again\)](#)

<https://www.youtube.com/watch?v=LCEmiRjPEtQ>

■ **Blog** 2025 [Anthropic: Claude Code best practices](#)

<https://www.anthropic.com/engineering/claude-code-best-practices>

Spec-driven development (SDD)

ACTIVE 2026 Since 2025–2026 Intermediate Production practice

Write a structured spec and plan first, then let agents implement against it, keeping the spec as source of truth.

- Typical flow: specify, plan, break into tasks, implement, verify.
- The spec doubles as durable context for future agent sessions.

■ **GitHub** 2025 [GitHub Spec Kit](#)

<https://github.com/github/spec-kit>

■ **Docs** 2025 [Kiro](#)

<https://kiro.dev>

Ralph loop (Ralph Wiggum technique)

ACTIVE 2026 Since 2025–2026 Intermediate

Running a coding agent in a simple while-loop against the same prompt and plan file until the job is done.

- Progress lives in files and git, not in the context window.
- Works best with strong tests so each iteration can check its own work.

■ **Blog** 2025 [Geoffrey Huntley: Ralph](#)

<https://ghuntley.com/ralph/>

Hooks, plugins and slash commands

ACTIVE 2026 Since 2025–2026 Intermediate Production practice

Extension points in coding agents: hooks run deterministic scripts at lifecycle events; plugins bundle commands, subagents, skills, hooks and MCP servers.

- Hooks enforce rules deterministically (formatters, linters, blocked commands).
- Plugins make a team's setup installable in one step.

■ **Docs** 2025 [Claude Code hooks](#)

<https://docs.claude.com/en/docs/claude-code/hooks>

■ **Docs** 2025 [Claude Code plugins](#)

<https://docs.claude.com/en/docs/claude-code/plugins>

Parallel agents with git worktrees

ACTIVE 2026 Since 2025–2026 Intermediate Production practice

Run several agents at once, each in its own worktree or container, then review and merge.

- Each agent gets its own branch and working directory, avoiding conflicts.
- The human shifts from typing to dispatching and reviewing.

■ **Docs** 2015 [git worktree docs](#)

<https://git-scm.com/docs/git-worktree>

■ **GitHub** 2025 [container-use](#)

<https://github.com/dagger/container-use>

The review bottleneck

NEW 2026 Since 2026 Intermediate Production practice

When code generation gets cheap, human review becomes the constraint. Changes grow larger and harder to review, and plausible-looking AI code hides subtle bugs.

- Mitigations: agent reviewers, architecture rules as tests, smaller PRs, agents that exercise the app themselves.

■ **Blog** 2026 [Faros: Harness engineering and the senior engineer tax](https://www.faros.ai/blog/harness-engineering)
<https://www.faros.ai/blog/harness-engineering>

5

Context, Memory and Retrieval

What the model sees on each call is the main lever you control at inference time: choosing, compressing and fetching the right information.

Context engineering

ACTIVE 2026 Since 2025–2026 Beginner Foundational, Production practice

Curating the smallest set of high-signal tokens (instructions, tools, examples, retrieved data, history) that the model needs for its next step.

- Successor to prompt engineering for agents.
- Techniques: just-in-time retrieval, compaction, structured notes, subagents with clean context.

- **Blog** 2025 [Anthropic: Effective context engineering for AI agents](https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents)
<https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>
- **Blog** 2025 [LangChain: Context engineering for agents](https://blog.langchain.com/context-engineering-for-agents/)
<https://blog.langchain.com/context-engineering-for-agents/>

Context rot

Since 2025 Beginner Research frontier

Accuracy degrades as input length grows, even well within the advertised window. Long context is not free.

- Distractors and similar-but-wrong passages hurt more than raw length.
- Mitigate with retrieval, summarisation and subagents.

STATE OF THE ART, SEPT 2026

Million-token windows are standard in 2026, but focused context still usually beats more context.

- **Blog** 2025 [Chroma Research: Context rot](https://research.trychroma.com/context-rot)
<https://research.trychroma.com/context-rot>

Compaction and context editing

ACTIVE 2026 Since 2025–2026 Intermediate Production practice

Summarising or pruning old turns and stale tool results as a conversation nears the limit, so agents can keep working indefinitely.

- Clear old tool outputs first; they are usually the largest and least useful.
- Keep decisions, open problems and file paths in the summary.

- **Blog** 2025 [Anthropic: Managing context on the Claude Developer Platform](https://www.anthropic.com/news/context-management)
<https://www.anthropic.com/news/context-management>

Prompt caching

ACTIVE 2026 Since 2024–2026 Intermediate Cost & efficiency, Production practice

Reusing the computed KV cache for a repeated prefix across requests to cut latency and cost.

- Harnesses keep prefixes stable (tool order, system prompt) so caches keep hitting.

STATE OF THE ART, SEPT 2026

Cache-read prices were cut steeply again in 2026, making long agent sessions much cheaper.

- **Docs** 2024 [Claude prompt caching](#)

<https://docs.claude.com/en/docs/build-with-claude/prompt-caching>

Agent memory

ACTIVE 2026 Since 2023–2026 Intermediate Production practice, Research frontier

Persisting facts, preferences and lessons beyond one context window: files the agent writes, vector or graph stores, or self-editing memory blocks.

- Memory quality, provenance and forgetting are now evaluated directly.
- Related to continual learning, but done outside the weights.

■ **GitHub** 2023 [Letta \(MemGPT\)](#)
<https://github.com/letta-ai/letta>

■ **GitHub** 2024 [mem0](#)
<https://github.com/mem0ai/mem0>

■ **GitHub** 2024 [Graphiti](#)
<https://github.com/getzep/graphiti>

Retrieval-augmented generation (RAG)

ACTIVE 2026 Since 2020–2026 Beginner Foundational, Production practice

Fetch relevant documents and put them in context before generating.

- Modern stacks: hybrid BM25 + vector search, rerankers, contextual chunk headers, query rewriting.

■ **Paper** 2020 [Original RAG paper](#)
<https://arxiv.org/abs/2005.11401>

■ **Blog** 2024 [Anthropic: Contextual retrieval](#)
<https://www.anthropic.com/news/contextual-retrieval>

■ **GitHub** 2022 [LlamaIndex](#)
https://github.com/run-llama/llama_index

Agentic search and deep research

ACTIVE 2026 Since 2025–2026 Intermediate Production practice

The agent plans queries, reads pages, follows leads and synthesises a cited report over many minutes.

- Many teams prefer agentic grep/file search over vector RAG for code.

■ **Blog** 2025 [OpenAI: Introducing deep research](#)
<https://openai.com/index/introducing-deep-research/>

■ **GitHub** 2023 [GPT Researcher](#)
<https://github.com/assafelovic/gpt-researcher>

GraphRAG

Since 2024–2025 Intermediate

Build a knowledge graph from a corpus, then retrieve over communities and paths to answer global questions.

- Entity and relation extraction with an LLM, then community summaries.
- More expensive to index than vector RAG.

■ **GitHub** 2024 [Microsoft GraphRAG](#)
<https://github.com/microsoft/graphrag>

■ **GitHub** 2024 [LightRAG](#)
<https://github.com/HKUDS/LightRAG>

Embeddings, late interaction and Matryoshka

Since 2020–2025 Intermediate

Dense embeddings map text to vectors; late-interaction models keep per-token vectors; Matryoshka embeddings can be truncated with little loss.

- Choose embeddings by your domain and language, not only leaderboard rank.
- Rerankers usually give the biggest quality jump after the first retrieval.

■ **Docs** 2022 [MTEB leaderboard](#)

<https://huggingface.co/spaces/mteb/leaderboard>

■ **GitHub** 2020 [ColBERT](#)

<https://github.com/stanford-futuredata/ColBERT>

■ **Paper** 2022 [Matryoshka representation learning](#)

<https://arxiv.org/abs/2205.13147>

Structured outputs and constrained decoding

Since 2023–2025 Intermediate Production practice

Force output to match a JSON schema or grammar by masking invalid tokens during decoding.

- Grammar engines compile schemas into token masks for near-zero overhead.
- Guarantees valid syntax, not correct content.

■ **GitHub** 2024 [XGrammar](#)

<https://github.com/mlc-ai/xgrammar>

■ **GitHub** 2023 [Outlines](#)

<https://github.com/dottxt-ai/outlines>

■ **Docs** 2024 [OpenAI structured outputs](#)

<https://platform.openai.com/docs/guides/structured-outputs>

6

Reasoning and Post-Training

Most capability gains since late 2024 came from post-training. In 2026 the recipe shifted again: many RL specialists, merged by on-policy distillation.

Reasoning models and test-time compute

ACTIVE 2026 Since 2024–2026 Beginner Foundational

Models trained to think in long chains before answering; accuracy scales with thinking tokens, a second scaling axis beyond model size.

- Techniques: long chain-of-thought, self-verification, parallel sampling with voting.
- More thinking helps most on math, code and multi-step planning.

STATE OF THE ART, SEPT 2026

In 2026 general and reasoning lines have merged: one model decides how much to think per request.

- **Blog** 2024 OpenAI: Learning to reason with LLMs
<https://openai.com/index/learning-to-reason-with-llms/>
- **Paper** 2024 Scaling test-time compute optimally
<https://arxiv.org/abs/2408.03314>
- **Paper** 2025 DeepSeek-R1
<https://arxiv.org/abs/2501.12948>

Thinking budgets, adaptive thinking and effort

ACTIVE 2026 Since 2025–2026 Intermediate Cost & efficiency, Production practice

Controls that set how much a model thinks per request, trading latency and cost for quality. Interleaved thinking lets it reason between tool calls.

- Low effort for chat and simple tool calls; high for hard debugging or proofs.
- Interleaved thinking helps agents adjust after each tool result.

STATE OF THE ART, SEPT 2026

Adjustable reasoning effort inside a conversation became a standard API feature in 2026.

- **Docs** 2025 Claude extended thinking docs
<https://docs.claude.com/en/docs/build-with-claude/extended-thinking>
- **Paper** 2025 s1: simple test-time scaling
<https://arxiv.org/abs/2501.19393>

RL with verifiable rewards (RLVR)

ACTIVE 2026 Since 2024–2026 Intermediate Foundational

Reinforcement learning where rewards come from checkers (tests, answer matching, verifiers) instead of a learned preference model.

- Coined in the Tulu 3 work (2024), popularised by DeepSeek-R1.
- RL is now a large and growing share of frontier training compute.

- **Paper** 2024 Tulu 3
<https://arxiv.org/abs/2411.15124>
- **Docs** 2024 RLHF Book (Nathan Lambert)
<https://rlhfbook.com>

Multi-teacher on-policy distillation (MOPD)

NEW 2026 Since 2026 Advanced Research frontier

The 2026 post-training pattern: train several domain specialists with SFT then RL, then train one general student on its own samples while minimising reverse-KL to the relevant teacher token by token.

- Avoids capability trade-offs from mixing math, code and agentic RL in one run.
- Specialists can be built in parallel by different teams.
- Introduced with Xiaomi MiMo, scaled to 10+ teachers by DeepSeek V4 and Nemotron 3 Ultra.

STATE OF THE ART, SEPT 2026

Now the default final stage in several frontier open recipes.

■ **Docs** 2026 [RLHF Book course: Frontier post-training recipe survey](https://rlhfbook.com/teach/course/conversation-01/)
<https://rlhfbook.com/teach/course/conversation-01/>

■ **Blog** 2026 [Hugging Face: Distillation in 2026](https://huggingface.co/blog/sergiopaniago/distillation-2026)
<https://huggingface.co/blog/sergiopaniago/distillation-2026>

■ **Blog** 2025 [Thinking Machines: On-policy distillation](https://thinkingmachines.ai/blog/on-policy-distillation/)
<https://thinkingmachines.ai/blog/on-policy-distillation/>

GRPO and its successors (GRPO, DAPO, GSPO)

Since 2024–2025 Advanced Research frontier

Group Relative Policy Optimization drops PPO's value network and scores samples against their group. Successors fix length bias, clipping and MoE instability.

- Sample a group of answers per prompt; advantage = reward minus group mean.
- No critic network means less memory and simpler training.

■ **Paper** 2024 [DeepSeekMath \(GRPO\)](https://arxiv.org/abs/2402.03300)
<https://arxiv.org/abs/2402.03300>

■ **Paper** 2025 [DAPO](https://arxiv.org/abs/2503.14476)
<https://arxiv.org/abs/2503.14476>

■ **Paper** 2025 [GSPO](https://arxiv.org/abs/2507.18071)
<https://arxiv.org/abs/2507.18071>

■ **Paper** 2025 [Dr. GRPO](https://arxiv.org/abs/2503.20783)
<https://arxiv.org/abs/2503.20783>

RL environments and verifiers

ACTIVE 2026 Since 2025–2026 Advanced Research frontier, Enterprise

Packaged tasks plus graders used to train agents. In 2026 environments look like sandboxed replicas of enterprise software, and verifier quality is seen as the next ceiling on agent progress.

- Labs spend heavily on environments; a market of environment vendors has formed.
- As agents improve, verifiers must improve faster to avoid reward hacking.

■ **GitHub** 2025 [Prime Intellect verifiers](https://github.com/PrimeIntellect-ai/verifiers)
<https://github.com/PrimeIntellect-ai/verifiers>

■ **GitHub** 2025 [OpenEnv](https://github.com/meta-pytorch/OpenEnv)
<https://github.com/meta-pytorch/OpenEnv>

■ **GitHub** 2024 [verl](https://github.com/volcengine/verl)
<https://github.com/volcengine/verl>

■ **GitHub** 2025 [SkyRL](https://github.com/NovaSky-AI/SkyRL)
<https://github.com/NovaSky-AI/SkyRL>

■ **Blog** 2026 [Prolific: RL environments for agentic AI \(2026 guide\)](https://www.prolific.com/resources/rl-environments-for-agentic-ai-the-2026-guide-to-building-and-deploying-enterprise-rl-environments)
<https://www.prolific.com/resources/rl-environments-for-agentic-ai-the-2026-guide-to-building-and-deploying-enterprise-rl-environments>

RLHF, DPO and preference optimization

Since 2022–2025 Intermediate Foundational

Aligning outputs to human preferences with a reward model plus RL, or directly from preference pairs (DPO).

- RLHF: SFT, then reward model, then PPO.
- DPO optimises the same objective in closed form from preference pairs.

■ **Paper** 2022 [InstructGPT](https://arxiv.org/abs/2203.02155)
<https://arxiv.org/abs/2203.02155>

■ **Paper** 2023 [DPO](https://arxiv.org/abs/2305.18290)
<https://arxiv.org/abs/2305.18290>

■ **GitHub** 2020 [Hugging Face TRL](https://github.com/huggingface/trl)
<https://github.com/huggingface/trl>

Constitutional AI and model specs

ACTIVE 2026 Since 2022–2026 Intermediate

Training behaviour from a written set of principles with AI feedback. Labs now publish the documents that define model character and priorities.

- The model critiques and revises its own outputs against principles.
- Published specs make intended behaviour inspectable.

■ **Paper** 2022 [Constitutional AI](https://arxiv.org/abs/2212.08073)
<https://arxiv.org/abs/2212.08073>

■ **Docs** 2026 [Claude’s constitution](https://www.anthropic.com/constitution)
<https://www.anthropic.com/constitution>

Process reward models (PRM)

Since 2023–2025 Advanced Research frontier

Reward models that grade each reasoning step rather than only the final answer.

- Enable step-level search (beam search over reasoning steps).
- Expensive to label; often trained with automated step checks.

■ **Paper** 2023 [Let’s verify step by step](https://arxiv.org/abs/2305.20050)
<https://arxiv.org/abs/2305.20050>

Reward hacking

ACTIVE 2026 Since 2016–2026 Intermediate Security, Research frontier

The policy exploits flaws in the reward (special-casing tests, editing graders) instead of solving the task; learned hacking can generalise to broader misbehaviour.

- Watch for tests being deleted, asserts weakened or outputs hard-coded.
- Defenses: held-out verifiers, monitoring, diverse environments.

■ **Blog** 2024 [Lilian Weng: Reward hacking](https://lilianweng.github.io/posts/2024-11-28-reward-hacking/)
<https://lilianweng.github.io/posts/2024-11-28-reward-hacking/>

■ **Blog** 2025 [Anthropic: Emergent misalignment from reward hacking](https://www.anthropic.com/research/emergent-misalignment-reward-hacking)
<https://www.anthropic.com/research/emergent-misalignment-reward-hacking>

Self-play and self-generated curricula

Since 2025 Advanced Research frontier

Models propose tasks, solve them and learn from verified results.

- A proposer creates tasks near the solver’s frontier.
- Verification (code execution, math checkers) keeps it grounded.

■ **Paper** 2025 [Absolute Zero](https://arxiv.org/abs/2505.03335)
<https://arxiv.org/abs/2505.03335>

Continual learning

ACTIVE 2026 Since 2024–2026 Advanced Research frontier

Updating a deployed model with new knowledge and skills without catastrophic forgetting.

- Approaches: replay, regularisation, architectural expansion, self-distillation, nested optimisation.
- Widely named as a key 2026 frontier for agents that improve on the job.

■ **Paper** 2026 **Survey: Continual learning in LLMs (2026)**

<https://arxiv.org/abs/2603.12658>

■ **Blog** 2025 **Google Research: Nested learning**

<https://research.google/blog/introducing-nested-learning-a-new-ml-paradigm-for-continual-learning/>

7

Model Architectures

Transformers still dominate, but 2026 models mix sparse experts, compressed and sparse attention, hybrid linear layers and new residual schemes.

Mixture of Experts (MoE)

ACTIVE 2026 Since 2017–2026 Beginner Foundational, Cost & efficiency

Tokens route to a few expert feed-forward blocks out of many, so total parameters grow while compute per token stays small.

- Fine-grained experts plus a shared expert is the common 2026 layout.
- Expert parallelism needs fast all-to-all communication.

STATE OF THE ART, SEPT 2026

Frontier open models reach 1.6T to 2.8T total parameters with roughly 3% active per token.

■ **Blog** 2023 [Hugging Face: MoE explained](https://huggingface.co/blog/moe)
<https://huggingface.co/blog/moe>

■ **Paper** 2024 [DeepSeek-V3](https://arxiv.org/abs/2412.19437)
<https://arxiv.org/abs/2412.19437>

■ **GitHub** 2025 [DeepEP](https://github.com/deepseek-ai/DeepEP)
<https://github.com/deepseek-ai/DeepEP>

GQA and Multi-head Latent Attention (GQA, MLA)

Since 2023–2025 Intermediate Cost & efficiency

Two ways to shrink the KV cache: share keys/values across head groups (GQA) or compress them into a low-rank latent (MLA).

- GQA: fewer KV heads than query heads.
- MLA: store a small latent, up-project at attention time; decoupled RoPE dims.

■ **Paper** 2023 [GQA](https://arxiv.org/abs/2305.13245)
<https://arxiv.org/abs/2305.13245>

■ **Paper** 2024 [DeepSeek-V2 \(MLA\)](https://arxiv.org/abs/2405.04434)
<https://arxiv.org/abs/2405.04434>

Sparse and compressed attention (NSA, DSA, CSA, HCA)

ACTIVE 2026 Since 2025–2026 Advanced Research frontier, Cost & efficiency

Attend only to selected tokens or compressed summaries of blocks to make long context cheap.

- DeepSeek V4 interleaves Compressed Sparse Attention (compress KV along the sequence, then top-k select) with Heavily Compressed Attention.
- Reported KV-cache reductions near 98% versus standard GQA make 1M-token inference affordable.

STATE OF THE ART, SEPT 2026

DeepSeek V4's CSA/HCA hybrid replaced MLA and is the most detailed public long-context design in 2026.

■ **Paper** 2025 [Native Sparse Attention](https://arxiv.org/abs/2502.11089)
<https://arxiv.org/abs/2502.11089>

■ **GitHub** 2025 [DeepSeek-V3.2-Exp \(DSA\)](https://github.com/deepseek-ai/DeepSeek-V3.2-Exp)
<https://github.com/deepseek-ai/DeepSeek-V3.2-Exp>

- **Paper** 2026 [DeepSeek-V4 report](https://arxiv.org/abs/2606.19348)
<https://arxiv.org/abs/2606.19348>

Manifold-constrained hyper-connections (mHC)

NEW 2026 Since 2026 Advanced Research frontier

Replaces plain residual connections with learned mixtures over several parallel residual streams, projected onto a constrained manifold to keep signals stable in very deep, very large models.

- Used in DeepSeek V4 to stabilise training at trillion-parameter scale.

- **Paper** 2026 [DeepSeek-V4 report](https://arxiv.org/abs/2606.19348)
<https://arxiv.org/abs/2606.19348>

Hybrid linear-attention and state space models (Mamba, Gated DeltaNet)

ACTIVE 2026 Since 2023–2026 Advanced Research frontier, Cost & efficiency

Mix a few full-attention layers with many linear-time layers for long context at lower cost.

- Linear layers keep a fixed-size state instead of a growing KV cache.
- A few full-attention layers preserve precise recall.

STATE OF THE ART, SEPT 2026

A leading 2026 trend: Qwen 3.x, Nemotron 3, Kimi Linear and others ship hybrids.

- **GitHub** 2023 [Mamba](https://github.com/state-spaces/mamba)
<https://github.com/state-spaces/mamba>
- **GitHub** 2024 [Gated DeltaNet](https://github.com/NVlabs/GatedDeltaNet)
<https://github.com/NVlabs/GatedDeltaNet>
- **GitHub** 2025 [Kimi Linear](https://github.com/MoonshotAI/Kimi-Linear)
<https://github.com/MoonshotAI/Kimi-Linear>
- **GitHub** 2023 [flash-linear-attention](https://github.com/fla-org/flash-linear-attention)
<https://github.com/fla-org/flash-linear-attention>

Diffusion language models (dLLM)

ACTIVE 2026 Since 2025–2026 Advanced Research frontier, Cost & efficiency

Generate text by iteratively denoising blocks in parallel rather than left to right, for much faster decoding.

- Start from masked or noisy tokens and refine in parallel steps.
- Trade-offs: quality on long reasoning, variable-length output.

STATE OF THE ART, SEPT 2026

Commercial diffusion LLMs now offer very low-cost, high-speed endpoints.

- **GitHub** 2025 [LLaDA](https://github.com/ML-GSAI/LLaDA)
<https://github.com/ML-GSAI/LLaDA>
- **Docs** 2025 [Gemini Diffusion](https://deepmind.google/models/gemini-diffusion/)
<https://deepmind.google/models/gemini-diffusion/>
- **Docs** 2025 [Inception Labs \(Mercury\)](https://www.inceptionlabs.ai)
<https://www.inceptionlabs.ai>

Multi-token prediction and speculative decoding (MTP)

ACTIVE 2026 Since 2023–2026 Intermediate Cost & efficiency

Predict several tokens at once, or let a cheap draft propose tokens that the big model verifies in one pass.

- Draft models, extra prediction heads (MTP, Medusa, EAGLE) or n-gram lookup.
- Output distribution is unchanged when verification is exact.

- **Paper** 2022 [Speculative decoding](https://arxiv.org/abs/2211.17192)
<https://arxiv.org/abs/2211.17192>

■ **GitHub** 2024 [EAGLE](#)

<https://github.com/SafeAILab/EAGLE>

■ **Paper** 2024 [Multi-token prediction](#)

<https://arxiv.org/abs/2404.19737>

Long-context methods (RoPE, YaRN)

ACTIVE 2026 Since 2021–2026 Intermediate

Rotary embeddings and scaling tricks that extend context windows; combined with sparse attention and KV compression to reach 1M tokens.

- Train short, extend with RoPE scaling plus long-context fine-tuning.
- Needle tests are easy; multi-hop reasoning over long input is hard.

STATE OF THE ART, SEPT 2026

1M-token context is the default for 2026 frontier models.

■ **Paper** 2021 [RoFormer \(RoPE\)](#)

<https://arxiv.org/abs/2104.09864>

■ **GitHub** 2023 [YaRN](#)

<https://github.com/jquesnelle/yarn>

Attention sinks

Since 2023–2025 Advanced Research frontier

Models park attention on the first tokens; keeping them (or learned sink logits) stabilises long generation.

- Keep the first few tokens when using sliding-window caches.
- Some models add explicit sink parameters per head.

■ **Paper** 2023 [StreamingLLM](#)

<https://arxiv.org/abs/2309.17453>

Tokenizer-free and byte-level models (BLT)

Since 2024–2025 Advanced Research frontier

Operate on bytes grouped into dynamic patches instead of a fixed vocabulary.

- Patch boundaries adapt to entropy, spending compute on hard bytes.
- Improves robustness on typos, code and rare scripts.

■ **GitHub** 2024 [Byte Latent Transformer](#)

<https://github.com/facebookresearch/blt>

Tiny recursive reasoning models (HRM, TRM)

Since 2025 Advanced Research frontier

Very small networks that loop over latent state, scoring well on puzzles like ARC-AGI with a few million parameters.

- Iterate a small network many times on its own latent answer.
- Shows depth-through-recursion can substitute for parameters on some tasks.

■ **GitHub** 2025 [Hierarchical Reasoning Model](#)

<https://github.com/sapientinc/HRM>

■ **GitHub** 2025 [Tiny Recursive Models](#)

<https://github.com/SamsungSAILMontreal/TinyRecursiveModels>

Small language models (SLM)

ACTIVE 2026 Since 2024–2026 Beginner Cost & efficiency, Open source / open weights

Compact models (about 0.5B to 30B) for on-device use, low latency and cheap agent subtasks.

- Often distilled from large teachers.
- Good fits: routing, extraction, classification, on-device assistants.

STATE OF THE ART, SEPT 2026

2026 mid-size open models (around 27B) deliver frontier-like coding on a single consumer GPU.

■ **Blog** 2025 [Hugging Face: SmolLM3](https://huggingface.co/blog/smollm3)
<https://huggingface.co/blog/smollm3>

■ **Docs** 2025 [Apple Foundation Models framework](https://developer.apple.com/documentation/foundationmodels)
<https://developer.apple.com/documentation/foundationmodels>

Architecture overviews

ACTIVE 2026 Since 2025–2026 Beginner

Side-by-side walkthroughs of how current open models differ.

- Compare norms, attention type, expert counts, positional schemes.
- Read tech reports for data and training details that diagrams omit.

■ **Blog** 2025 [Raschka: The big LLM architecture comparison](https://magazine.sebastianraschka.com/p/the-big-llm-architecture-comparison)
<https://magazine.sebastianraschka.com/p/the-big-llm-architecture-comparison>

■ **Blog** 2026 [Raschka: LLM research papers 2026](https://magazine.sebastianraschka.com/p/llm-research-papers-2026-part1)
<https://magazine.sebastianraschka.com/p/llm-research-papers-2026-part1>

8

Training at Scale and Fine-Tuning

How models are pretrained across thousands of accelerators, and how practitioners adapt them cheaply afterwards.

Parallelism strategies (DP, FSDP, TP, PP, EP, CP)

ACTIVE 2026 Since 2019–2026 Advanced Foundational

Data, fully-sharded, tensor, pipeline, expert and context parallelism split models and data across devices; real runs combine several.

- FSDP shards parameters, gradients and optimizer state.
- Tensor and expert parallelism need fast scale-up links; pipeline tolerates slower links.

■ **Blog** 2025 [HF: The Ultra-Scale Playbook](https://huggingface.co/spaces/nanotron/ultrascale-playbook)
<https://huggingface.co/spaces/nanotron/ultrascale-playbook>

■ **Blog** 2025 [How to Scale Your Model \(JAX\)](https://jax-ml.github.io/scaling-book/)
<https://jax-ml.github.io/scaling-book/>

■ **GitHub** 2024 [torchtititan](https://github.com/pytorch/torchtititan)
<https://github.com/pytorch/torchtititan>

■ **GitHub** 2019 [Megatron-LM](https://github.com/NVIDIA/Megatron-LM)
<https://github.com/NVIDIA/Megatron-LM>

■ **GitHub** 2020 [DeepSpeed](https://github.com/deepspeedai/DeepSpeed)
<https://github.com/deepspeedai/DeepSpeed>

Scaling laws

ACTIVE 2026 Since 2020–2026 Intermediate Foundational

Empirical power laws relating loss to parameters, data and compute; now also studied for RL and test-time compute.

- Compute-optimal data-to-parameter ratio was roughly 20 tokens per parameter (Chinchilla).
- Inference cost now pushes teams to over-train smaller models.

■ **Paper** 2022 [Chinchilla](https://arxiv.org/abs/2203.15556)
<https://arxiv.org/abs/2203.15556>

■ **Docs** 2022 [Epoch AI data](https://epoch.ai/data)
<https://epoch.ai/data>

Muon and modern optimizers

ACTIVE 2026 Since 2024–2026 Advanced Research frontier

Muon orthogonalises momentum updates for matrix parameters and trains faster than AdamW at scale.

- Newton-Schulz iterations approximate orthogonalisation cheaply.
- Usually combined with AdamW for embeddings and norms.

STATE OF THE ART, SEPT 2026

Adopted in frontier open models including Kimi K2 and DeepSeek V4.

■ **Blog** 2024 [Keller Jordan: Muon](https://kellerjordan.github.io/posts/muon/)
<https://kellerjordan.github.io/posts/muon/>

■ **GitHub** 2025 [Kimi K2](https://github.com/MoonshotAI/Kimi-K2)
<https://github.com/MoonshotAI/Kimi-K2>

■ **GitHub** 2024 [modded-nanogpt](https://github.com/KellerJordan/modded-nanogpt)
<https://github.com/KellerJordan/modded-nanogpt>

Data curation and synthetic data

ACTIVE 2026 Since 2023–2026 Intermediate

Classifier filtering, deduplication, and synthetic textbooks, reasoning traces and agent trajectories.

- Quality classifiers trained on curated seed data filter web crawls.
- Synthetic data risks collapse without real-data anchoring and verification.

STATE OF THE ART, SEPT 2026

DeepSeek V4 pretrained on 32T+ tokens, more than double V3's corpus.

■ **Blog** 2024 [FineWeb](https://huggingface.co/spaces/HuggingFaceFW/blogpost-fineweb-v1)
<https://huggingface.co/spaces/HuggingFaceFW/blogpost-fineweb-v1>

■ **Blog** 2024 [Cosmopedia](https://huggingface.co/blog/cosmopedia)
<https://huggingface.co/blog/cosmopedia>

Mid-training

ACTIVE 2026 Since 2024–2026 Intermediate

A stage between pretraining and post-training that shifts the mix toward code, math, long context and reasoning.

- Often includes long-context extension and instruction-like data.
- Sets up the base model for efficient RL later.

■ **Docs** 2024 [Allen AI OLMo](https://allenai.org/olmo)
<https://allenai.org/olmo>

LoRA and parameter-efficient fine-tuning (PEFT, LoRA, QLoRA)

Since 2021–2025 Intermediate Cost & efficiency

Train small low-rank adapters instead of all weights; QLoRA does it on a 4-bit base.

- Adapters are small files you can swap per customer or task.
- Higher ranks and all-layer adapters close most of the gap to full fine-tuning.

■ **Paper** 2021 [LoRA](https://arxiv.org/abs/2106.09685)
<https://arxiv.org/abs/2106.09685>

■ **Paper** 2023 [QLoRA](https://arxiv.org/abs/2305.14314)
<https://arxiv.org/abs/2305.14314>

■ **Blog** 2025 [Thinking Machines: LoRA without regret](https://thinkingmachines.ai/blog/loralora/)
<https://thinkingmachines.ai/blog/loralora/>

■ **GitHub** 2023 [HF PEFT](https://github.com/huggingface/peft)
<https://github.com/huggingface/peft>

Fine-tuning toolkits and APIs

ACTIVE 2026 Since 2023–2026 Beginner Open source / open weights

Tools that make SFT, DPO and RL fine-tuning accessible on one GPU or as a managed API.

- Start with SFT on a few thousand good examples before trying RL.
- Managed APIs hide distributed training but constrain algorithms.

■ **GitHub** 2023 [Unsloth](https://github.com/unslothai/unsloth)
<https://github.com/unslothai/unsloth>

■ **GitHub** 2023 [Axolotl](https://github.com/axolotl-ai-cloud/axolotl)
<https://github.com/axolotl-ai-cloud/axolotl>

■ **GitHub** 2023 [LLaMA-Factory](https://github.com/hiyouga/LLaMA-Factory)
<https://github.com/hiyouga/LLaMA-Factory>

■ **Docs** 2025 [Tinker](https://thinkingmachines.ai/tinker/)
<https://thinkingmachines.ai/tinker/>

Model merging

Since 2023–2025 Intermediate

Combining weights of several fine-tunes (SLERP, TIES, DARE) without further training.

- Works best between fine-tunes of the same base model.
- Popular for combining skills or languages cheaply.

■ **GitHub** 2023 [mergekit](https://github.com/arcee-ai/mergekit)
<https://github.com/arcee-ai/mergekit>

Decentralized training (DiLoCo)

ACTIVE 2026 Since 2023–2026 Advanced Research frontier

Training across loosely connected clusters that synchronise rarely.

- Workers train locally for many steps, then average updates.
- Enables training across datacenters and volunteer compute.

■ **Paper** 2023 [DiLoCo](https://arxiv.org/abs/2311.08105)
<https://arxiv.org/abs/2311.08105>

■ **GitHub** 2025 [prime-rl](https://github.com/PrimeIntellect-ai/prime-rl)
<https://github.com/PrimeIntellect-ai/prime-rl>

Build-it-yourself references

ACTIVE 2026 Since 2023–2026 Beginner Open source / open weights

Minimal end-to-end codebases for learning how chat models are trained.

- nanochat trains a small ChatGPT-like model end to end for a few hundred dollars.
- CS336 covers tokenizers, kernels, scaling and alignment from scratch.

■ **GitHub** 2025 [nanochat](https://github.com/karpathy/nanochat)
<https://github.com/karpathy/nanochat>

■ **Docs** 2025 [Stanford CS336](https://stanford-cs336.github.io)
<https://stanford-cs336.github.io>

9

Inference and Serving

Serving cost now rivals training cost. These techniques decide tokens per second, latency and price.

Inference engines

ACTIVE 2026 Since 2023–2026 Beginner Open source / open weights, Production practice

Servers that batch requests, manage the KV cache and run optimised kernels.

- Pick by hardware, model support and features (disaggregation, LoRA, structured output).
- Benchmark on your own traffic shape.

STATE OF THE ART, SEPT 2026

vLLM and SGLang lead open serving; llama.cpp, Ollama and MLX dominate local use.

- **GitHub 2023 vLLM**
<https://github.com/vllm-project/vllm>
- **GitHub 2024 SGLang**
<https://github.com/sgl-project/sglang>
- **GitHub 2023 TensorRT-LLM**
<https://github.com/NVIDIA/TensorRT-LLM>
- **GitHub 2023 llama.cpp**
<https://github.com/ggml-org/llama.cpp>
- **GitHub 2023 Ollama**
<https://github.com/ollama/ollama>
- **GitHub 2023 MLX**
<https://github.com/ml-explore/mlx>

KV cache and PagedAttention

Since 2023–2025 Intermediate Foundational

Keys and values of past tokens are cached; PagedAttention stores them in fixed pages like virtual memory.

- KV memory grows with batch size times sequence length.
- Paging enables sharing and near-zero waste.

- **Paper 2023 PagedAttention**
<https://arxiv.org/abs/2309.06180>

Continuous batching and prefix caching

Since 2022–2025 Intermediate Cost & efficiency

Add and remove requests every step, and share cached prefixes (radix trees) across requests.

- Keeps GPUs busy despite requests of different lengths.
- Prefix sharing is huge for agents with long common system prompts.

- **GitHub 2024 SGLang (RadixAttention)**
<https://github.com/sgl-project/sglang>

Prefill/decode disaggregation (PD disaggregation)

ACTIVE 2026 Since 2024–2026 Advanced Production practice, Cost & efficiency

Run compute-bound prefill and memory-bound decode on separate pools, moving KV caches between them.

- NVIDIA now builds separate rack types for prefill (CPX) and decode, scheduled together by Dynamo.

STATE OF THE ART, SEPT 2026

Standard for large-scale serving of million-token contexts in 2026.

■ Paper 2024 DistServe

<https://arxiv.org/abs/2401.09670>

■ GitHub 2025 NVIDIA Dynamo

<https://github.com/ai-dynamo/dynamo>

■ GitHub 2025 llm-d

<https://github.com/llm-d/llm-d>

■ GitHub 2024 Mooncake

<https://github.com/kvcache-ai/Mooncake>

KV cache offloading

ACTIVE 2026 Since 2024–2026 Advanced Cost & efficiency

Spill KV caches to CPU memory, SSD or remote storage and reload them for later turns.

- Turns re-prefill into a memory read.
- Tiered storage: HBM, CPU DRAM, local SSD, network.

■ GitHub 2024 LMCache

<https://github.com/LMCache/LMCache>

Attention kernels (FlashAttention)

ACTIVE 2026 Since 2022–2026 Advanced

IO-aware fused attention kernels that avoid materialising the full attention matrix.

- Tiling keeps data in fast on-chip SRAM.
- New versions target Hopper, Blackwell and FP8/FP4 paths.

■ GitHub 2022 FlashAttention

<https://github.com/Dao-AILab/flash-attention>

■ GitHub 2023 FlashInfer

<https://github.com/flashinfer-ai/flashinfer>

On-device and edge inference

ACTIVE 2026 Since 2023–2026 Beginner Cost & efficiency

Running models on phones, laptops and embedded hardware with quantised weights and NPU runtimes.

- Privacy and offline use are the main drivers.
- Memory bandwidth, not compute, limits phone and laptop token rates.

■ GitHub 2023 ExecuTorch

<https://github.com/pytorch/executorch>

■ GitHub 2018 ONNX Runtime

<https://github.com/microsoft/onnxruntime>

LLM gateways and routers

ACTIVE 2026 Since 2023–2026 Beginner Production practice, Cost & efficiency

One API over many providers with fallbacks, budgets, caching and routing to cheaper models.

- Central place for keys, rate limits, logging and cost control.
- Routers send easy prompts to cheap models.

■ **GitHub** 2023 [LiteLLM](https://github.com/BerriAI/litellm)
<https://github.com/BerriAI/litellm>

■ **Docs** 2023 [OpenRouter](https://openrouter.ai)
<https://openrouter.ai>

Inference benchmarking

ACTIVE 2026 Since 2025–2026 Intermediate Cost & efficiency

Measuring throughput, latency and cost per million tokens across hardware and software stacks.

- Report throughput at a fixed latency target.
- Cost per million tokens depends on batch size and cache hit rate.

■ **GitHub** 2025 [InferenceMAX](https://github.com/InferenceMAX/InferenceMAX)
<https://github.com/InferenceMAX/InferenceMAX>

■ **Docs** 2024 [Artificial Analysis](https://artificialanalysis.ai)
<https://artificialanalysis.ai>

10

Quantization and Number Formats

Lower precision is the cheapest speed-up available, and formats are increasingly standardised across vendors.

Microscaling formats (MX, MXFP8, MXFP4)

ACTIVE 2026 Since 2023–2026 Advanced Open standard, Cost & efficiency

An Open Compute Project standard: blocks of values share one scale, making 8-, 6- and 4-bit floats accurate enough for training and inference.

- Shared 8-bit exponent scale per 32-element block.
- Supported by NVIDIA, AMD, Intel, Arm, Meta and Microsoft.

■ **Spec** 2023 OCP Microscaling (MX) spec

<https://www.opencompute.org/documents/ocp-microscaling-formats-mx-v1-0-spec-final-pdf>

■ **Paper** 2023 Microscaling data formats

<https://arxiv.org/abs/2310.10537>

■ **GitHub** 2025 openai/gpt-oss (MXFP4)

<https://github.com/openai/gpt-oss>

NVFP4

ACTIVE 2026 Since 2025–2026 Advanced Cost & efficiency

NVIDIA's 4-bit float with 16-value blocks and FP8 scales, native on Blackwell and Rubin.

- Smaller blocks and finer scales than MXFP4 improve accuracy.
- Increasingly used for pretraining as well as inference.

STATE OF THE ART, SEPT 2026

Rubin rack performance is quoted primarily in FP4 exaflops.

■ **Blog** 2025 NVIDIA: Introducing NVFP4

<https://developer.nvidia.com/blog/introducing-nvfp4-for-efficient-and-accurate-low-precision-inference/>

FP8 training and inference (E4M3, E5M2)

ACTIVE 2026 Since 2022–2026 Intermediate

Two 8-bit float layouts used for large-scale training and routine serving; KV caches are often stored in FP8.

- E4M3 for weights and activations, E5M2 for gradients.
- Needs per-tensor or per-block scaling to avoid overflow.

■ **Paper** 2022 FP8 formats for deep learning

<https://arxiv.org/abs/2209.05433>

Post-training quantization (GPTQ, AWQ)

Since 2022–2025 Intermediate Cost & efficiency

Compress trained weights to 4 or 8 bits using calibration data.

- 4-bit weights with 16-bit activations is the common local setup.
- Always re-run your evals after quantizing.

■ **Paper** 2022 GPTQ

<https://arxiv.org/abs/2210.17323>

■ **GitHub** 2023 [AWQ](#)<https://github.com/mit-han-lab/llm-awq>■ **GitHub** 2024 [llm-compressor](#)<https://github.com/vllm-project/llm-compressor>

Quantization-aware training (QAT)

ACTIVE 2026 Since 2024–2026 Advanced

Simulate low precision during training so the model tolerates it.

- Fake-quantize in the forward pass, keep full-precision master weights.
- Recovers much of the accuracy lost to aggressive PTQ.

■ **GitHub** 2023 [torchao](#)<https://github.com/pytorch/ao>

GGUF and local model formats

ACTIVE 2026 Since 2023–2026 Beginner Open source / open weights

A single-file format for quantised models used by llama.cpp and local runners.

- K-quants (Q4_K_M, Q5_K_M) balance size and quality.
- Metadata stores tokenizer and chat template in the file.

■ **Spec** 2023 [GGUF specification](#)<https://github.com/ggml-org/ggml/blob/master/docs/gguf.md>■ **GitHub** 2021 [bitsandbytes](#)<https://github.com/bitsandbytes-foundation/bitsandbytes>

Ternary and 1-bit models (BitNet)

Since 2024–2025 Advanced Research frontier

Weights restricted to -1, 0, +1, trained from scratch for very cheap CPU inference.

- Matrix multiplies become additions and subtractions.
- Must be trained low-bit from the start.

■ **GitHub** 2024 [Microsoft BitNet](#)<https://github.com/microsoft/BitNet>

11

AI Hardware, Interconnect and Datacenter Standards

The unit of compute is now a rack or pod. Open standards are forming to connect accelerators from different vendors.

Rack-scale systems (NVL72)

ACTIVE 2026 Since 2024–2026 Intermediate

Dozens of accelerators in one liquid-cooled rack joined by a single scale-up domain, behaving like one giant GPU.

- GB200/GB300 NVL72 (Blackwell) dominated 2025.
- Vera Rubin NVL72 (72 Rubin GPUs, 36 Vera CPUs, NVLink 6, HBM4) began shipping to major clouds in 2026.

STATE OF THE ART, SEPT 2026

NVIDIA quotes about 3.6 EF FP4 inference per Vera Rubin rack; Rubin Ultra (Kyber rack) is planned for H2 2027.

■ Docs 2024 NVIDIA GB200 NVL72

<https://www.nvidia.com/en-us/data-center/gb200-nvl72/>

■ Blog 2026 ServeTheHome: Rubin platform at CES 2026

<https://www.servethehome.com/nvidia-launches-next-generation-rubin-ai-compute-platform-at-ces-2026/>

■ Blog 2026 DCD: Vera Rubin superchip

<https://www.datacenterdynamics.com/en/news/nvidia-announces-vera-rubin-superchip-for-late-2026/>

AMD Helios and Open Rack Wide (MI455X)

NEW 2026 Since 2026 Intermediate Open standard

AMD's rack-scale platform built on the Open Rack Wide standard co-submitted with Meta to OCP, pairing MI455X GPUs (432 GB HBM4 each) with EPYC Venice CPUs.

- Competes on memory capacity rather than raw FLOPs.
- Large 2026 deployments announced with OpenAI and Oracle.

■ Docs 2023 AMD Instinct

<https://www.amd.com/en/products/accelerators/instinct.html>

■ Blog 2026 GPU Insights: Rubin vs Helios

<https://gpuinsights.net/nvidia-rubin-vs-amd-helios-mi450-2026/>

Scale-up vs. scale-out networking

ACTIVE 2026 Since 2024–2026 Intermediate Foundational

Scale-up links accelerators inside a pod with memory-semantic, sub-microsecond links; scale-out connects pods over Ethernet or InfiniBand.

- Scale-up carries tensor and expert parallel traffic.
- Scale-out carries data-parallel gradients and inter-pod traffic.

■ Blog 2026 Synopsys: Ethernet standards for scale-up AI

<https://www.synopsys.com/articles/ethernet-standards-scale-up-ai.html>

UALink (Ultra Accelerator Link)

ACTIVE 2026 Since 2024–2026 Intermediate Open standard

An open memory-semantic interconnect for pods of up to 1,024 accelerators, backed by AMD, Google, Microsoft, Meta, AWS and 100+ members.

- UALink 200G 1.0 (April 2025); version 2.0 published April 2026.
- First UALink switches targeted late 2026.

■ **Spec** 2024 **UALink Consortium**
<https://ualinkconsortium.org>

■ **Docs** 2024 **UALink overview**
<https://en.wikipedia.org/wiki/UALink>

■ **Blog** 2026 **UALink white paper (2026)**
https://ualinkconsortium.org/wp-content/uploads/2026/01/UALink_White_Paper_Publication_Candidate_FINAL_VERSION.pdf

ESUN and SUE (Ethernet for Scale-Up Networking)

ACTIVE 2026 Since 2025–2026 Advanced Open standard

An OCP workstream (October 2025) standardising Ethernet’s lower layers for scale-up AI; Broadcom’s SUE evolved into a transport that sits on top.

- Backed by AMD, Arista, ARM, Broadcom, Cisco, HPE, Marvell, Meta, Microsoft, NVIDIA, OpenAI and Oracle.

■ **Blog** 2025 **Arista: The sun rises on scale-up Ethernet**
<https://blogs.arista.com/blog/the-sun-rises-on-scale-up-ethernet>

■ **Blog** 2026 **Synopsys: ESUN explained**
<https://www.synopsys.com/blogs/chip-design/esun-ethernet-ai-scale-up-networking.html>

■ **Blog** 2026 **650 Group: Ethernet in scale-up**
<https://650group.com/blog/in-the-ai-era-ethernet-set-to-surge-in-scale-out-and-ramp-in-scale-up/>

Ultra Ethernet (UEC)

ACTIVE 2026 Since 2023–2026 Advanced Open standard

A consortium spec modernising Ethernet transport (packet spraying, congestion control, telemetry) for scale-out AI.

- UEC 1.0 specification released in 2025.
- Targets InfiniBand-class performance on Ethernet.

■ **Spec** 2023 **Ultra Ethernet Consortium**
<https://ultraethernet.org>

NVLink and NVLink Fusion

ACTIVE 2026 Since 2016–2026 Intermediate

NVIDIA’s proprietary scale-up fabric, licensed through NVLink Fusion to third-party CPUs and accelerators.

- NVLink switches give all-to-all bandwidth inside a rack.
- Fusion opens the fabric to custom silicon partners.

STATE OF THE ART, SEPT 2026

NVLink 6 ships with Rubin.

■ **Docs** 2025 **NVLink Fusion**
<https://www.nvidia.com/en-us/data-center/nvlink-fusion/>

CXL, PCIe and UCIe

ACTIVE 2026 Since 2019–2026 Advanced Open standard

Coherent memory pooling (CXL), host I/O (PCIe) and chiplet die-to-die links (UCIe).

- CXL memory expansion adds capacity for KV caches and embeddings.
- UCIe lets vendors mix chiplets from different sources.

■ **Spec** 2019 **CXL Consortium**
<https://computeexpresslink.org>

■ **Spec** 1992 **PCI-SIG**
<https://pcisig.com>

■ **Spec** 2022 **UCIe**
<https://www.uciexpress.org>

High-bandwidth memory (HBM3E, HBM4)

ACTIVE 2026 Since 2013–2026 Intermediate

Stacked DRAM beside the accelerator; capacity and bandwidth usually limit decoding speed more than FLOPs.

- HBM4 doubles interface width versus HBM3.
- HBM supply is a key constraint on accelerator shipments.

STATE OF THE ART, SEPT 2026

HBM4 arrived in 2026 with Rubin and MI455X; HBM4E is on 2027 roadmaps.

■ **Docs** 2013 **High Bandwidth Memory overview**
https://en.wikipedia.org/wiki/High_Bandwidth_Memory

Advanced packaging (CoWoS)

ACTIVE 2026 Since 2012–2026 Advanced

2.5D/3D packaging that places compute dies and HBM on a shared interposer; a recurring supply bottleneck.

- Interposer size limits how much HBM fits beside a die.
- Packaging capacity expansions track AI demand closely.

■ **Docs** 2012 **TSMC CoWoS**
<https://3dfabric.tsmc.com/english/dedicatedFoundry/technology/cowos.htm>

Co-packaged optics (CPO)

ACTIVE 2026 Since 2025–2026 Advanced

Optical transceivers moved into the switch package to cut power and raise bandwidth.

- Shorter electrical paths mean lower power per bit.
- Early deployment in switches; accelerators later.

■ **Docs** 2025 **NVIDIA silicon photonics**
<https://www.nvidia.com/en-us/networking/products/silicon-photonics/>

Custom accelerators (TPU, Trainium, MTIA)

ACTIVE 2026 Since 2016–2026 Intermediate

Hyperscaler and challenger chips competing with GPUs, often optimised for inference.

- Hyperscalers design chips for their own inference fleets.
- Software ecosystem maturity is the main adoption hurdle.

■ **Blog** 2025 **Google: Ironwood TPU**
<https://blog.google/products/google-cloud/ironwood-tpu-age-of-inference/>

■ **Docs** 2020 **AWS Trainium**
<https://aws.amazon.com/ai/machine-learning/trainium/>

■ **Docs** 2016 [Cerebras](https://www.cerebras.ai)
<https://www.cerebras.ai>

■ **Docs** 2016 [Groq](https://groq.com)
<https://groq.com>

AI factories, power and cooling (800 VDC)

ACTIVE 2026 Since 2024–2026 Intermediate

Gigawatt-scale datacenters with direct liquid cooling and higher-voltage DC power; power is now a primary constraint.

- Racks exceed 100 kW; next-gen racks approach 600 kW.
- Grid access and power purchase deals shape site selection.

■ **Blog** 2025 [NVIDIA: 800 V HVDC architecture](https://developer.nvidia.com/blog/nvidia-800-v-hvdc-architecture-will-power-the-next-generation-of-ai-factories/)
<https://developer.nvidia.com/blog/nvidia-800-v-hvdc-architecture-will-power-the-next-generation-of-ai-factories/>

■ **Spec** 2011 [Open Compute Project](https://www.opencompute.org)
<https://www.opencompute.org>

■ **Blog** 2025 [OpenAI: Stargate](https://openai.com/index/announcing-the-stargate-project/)
<https://openai.com/index/announcing-the-stargate-project/>

Kernel languages and compilers (Triton, CUTLASS, ROCm, XLA)

ACTIVE 2026 Since 2019–2026 Advanced Open source / open weights

Tools for writing fast accelerator code without raw CUDA, and portable compilers.

- Triton makes GPU kernels writable in Python-like code.
- Portable stacks reduce lock-in to one vendor.

■ **GitHub** 2019 [Triton](https://github.com/triton-lang/triton)
<https://github.com/triton-lang/triton>

■ **GitHub** 2017 [CUTLASS / CuTe](https://github.com/NVIDIA/cutlass)
<https://github.com/NVIDIA/cutlass>

■ **GitHub** 2024 [ThunderKittens](https://github.com/HazyResearch/ThunderKittens)
<https://github.com/HazyResearch/ThunderKittens>

■ **GitHub** 2025 [TileLang](https://github.com/tile-ai/tilelang)
<https://github.com/tile-ai/tilelang>

■ **GitHub** 2016 [ROCm](https://github.com/ROCm/ROCm)
<https://github.com/ROCm/ROCm>

■ **GitHub** 2022 [OpenXLA](https://github.com/openxla/xla)
<https://github.com/openxla/xla>

■ **GitHub** 2023 [Modular \(Mojo, MAX\)](https://github.com/modular/modular)
<https://github.com/modular/modular>

Collective communication (NCCL)

ACTIVE 2026 Since 2016–2026 Advanced

Libraries for all-reduce, all-gather and all-to-all across GPUs.

- Topology-aware algorithms (rings, trees) matter at scale.
- MoE all-to-all is latency-sensitive.

■ **GitHub** 2016 [NCCL](https://github.com/NVIDIA/nccl)
<https://github.com/NVIDIA/nccl>

■ **GitHub** 2025 [DeepEP](https://github.com/deepseek-ai/DeepEP)
<https://github.com/deepseek-ai/DeepEP>

Model interchange formats (ONNX, safetensors)

Since 2017–2025 Beginner Open standard

Standard, safe formats for moving models between frameworks; safetensors avoids pickle code execution.

- ONNX targets cross-runtime deployment.
- safetensors is the default on Hugging Face.

■ GitHub 2017 ONNX

<https://github.com/onnx/onnx>

■ GitHub 2022 safetensors

<https://github.com/huggingface/safetensors>

Hardware benchmarks (MLPerf)

ACTIVE 2026 Since 2018–2026 Beginner

Industry-standard training and inference benchmarks.

- Compare systems on the same model and accuracy target.
- Vendor-submitted; read the fine print on configurations.

■ Docs 2018 MLCommons benchmarks

<https://mlcommons.org/benchmarks/>

■ Docs 2023 Epoch AI: ML hardware data

<https://epoch.ai/data/machine-learning-hardware>

12

Vision-Language and Multimodal Models

VLMs join a vision encoder to a language model so one system can read images, documents, screens and video. Omni models add audio in and out.

VLM architecture

ACTIVE 2026 Since 2023–2026 Beginner Foundational

A vision encoder turns image patches into tokens, a projector maps them into the LLM space, and the LLM reasons over both. Native multimodal models train everything jointly.

- Visual token count drives cost: compress or pool where possible.
- Training stages: alignment, instruction tuning, then RL.

■ **Blog** 2025 [Hugging Face: Vision language models 2025](https://huggingface.co/blog/vlms-2025)
<https://huggingface.co/blog/vlms-2025>

■ **GitHub** 2023 [LLaVA](https://github.com/haotian-liu/LLaVA)
<https://github.com/haotian-liu/LLaVA>

■ **Paper** 2025 [SigLIP 2](https://arxiv.org/abs/2502.14786)
<https://arxiv.org/abs/2502.14786>

Open VLM families

ACTIVE 2026 Since 2024–2026 Beginner Open source / open weights

Open-weight VLMs for images, documents and video, including small on-device models.

- Check licence, context length and video support.
- Evaluate on your own documents and images.

STATE OF THE ART, SEPT 2026

Closed models (Gemini, GPT, Claude) still top leaderboards; Qwen-VL, InternVL and Gemma 4 (April 2026, vision in every size) lead open options.

■ **GitHub** 2025 [Qwen3-VL](https://github.com/QwenLM/Qwen3-VL)
<https://github.com/QwenLM/Qwen3-VL>

■ **GitHub** 2023 [InternVL](https://github.com/OpenGVLab/InternVL)
<https://github.com/OpenGVLab/InternVL>

■ **Blog** 2024 [SmolVLM](https://huggingface.co/blog/smolvlm)
<https://huggingface.co/blog/smolvlm>

■ **Blog** 2026 [BentoML: Open VLM guide](https://www.bentoml.com/blog/multimodal-ai-a-guide-to-open-source-vision-language-models)
<https://www.bentoml.com/blog/multimodal-ai-a-guide-to-open-source-vision-language-models>

Native and dynamic resolution

Since 2023–2025 Intermediate

Handle images at their original aspect ratio and resolution by tiling or packing variable patch counts.

- Critical for small text, charts and screenshots.
- Token budget caps keep large images affordable.

■ **Paper** 2023 [NaViT](https://arxiv.org/abs/2307.06304)
<https://arxiv.org/abs/2307.06304>

Thinking with images

ACTIVE 2026 Since 2025–2026 Intermediate

Reasoning models that crop, zoom and transform images inside their chain of thought.

- Tool calls like crop and zoom inside reasoning.
- Improves fine-grained detail questions.

■ **Blog** 2025 [OpenAI: Thinking with images](https://openai.com/index/thinking-with-images/)
<https://openai.com/index/thinking-with-images/>

Document AI and VLM-based OCR

ACTIVE 2026 Since 2024–2026 Intermediate Production practice, Open source / open weights

VLMs that convert PDFs and scans into clean Markdown, tables and structure.

- Outputs Markdown with tables, formulas and reading order.
- Visual token compression can shrink long documents.

■ **GitHub** 2025 [DeepSeek-OCR](https://github.com/deepseek-ai/DeepSeek-OCR)
<https://github.com/deepseek-ai/DeepSeek-OCR>

■ **GitHub** 2025 [olmOCR](https://github.com/allenai/olmocr)
<https://github.com/allenai/olmocr>

■ **GitHub** 2024 [Docling](https://github.com/docling-project/docling)
<https://github.com/docling-project/docling>

■ **GitHub** 2020 [PaddleOCR](https://github.com/PaddlePaddle/PaddleOCR)
<https://github.com/PaddlePaddle/PaddleOCR>

Visual document retrieval (ColPali)

Since 2024–2025 Intermediate

Embed page images directly and retrieve with late interaction, skipping OCR.

- Handles charts, layouts and scans that OCR mangles.
- Storage cost is higher than single-vector embeddings.

■ **GitHub** 2024 [ColPali](https://github.com/illuin-tech/colpali)
<https://github.com/illuin-tech/colpali>

GUI grounding agents

ACTIVE 2026 Since 2024–2026 Advanced

VLMs trained to locate UI elements and act on screens.

- Output click coordinates or element IDs.
- Benchmarks: ScreenSpot, OSWorld.

■ **GitHub** 2025 [UI-TARS](https://github.com/bytedance/UI-TARS)
<https://github.com/bytedance/UI-TARS>

■ **GitHub** 2024 [OmniParser](https://github.com/microsoft/OmniParser)
<https://github.com/microsoft/OmniParser>

Omni and speech-to-speech models

ACTIVE 2026 Since 2024–2026 Intermediate

Models that take and produce audio natively for low-latency voice agents.

- End-to-end audio avoids ASR-LLM-TTS latency.
- Barge-in and turn detection are key UX problems.

STATE OF THE ART, SEPT 2026

2026 omni models handle text, image, video and audio input with 1M-token windows.

■ **GitHub** 2025 [Qwen3-Omni](https://github.com/QwenLM/Qwen3-Omni)
<https://github.com/QwenLM/Qwen3-Omni>

- **GitHub** 2024 [Moshi](https://github.com/kyutai-labs/moshi)
<https://github.com/kyutai-labs/moshi>
- **Docs** 2024 [OpenAI Realtime API](https://platform.openai.com/docs/guides/realtime)
<https://platform.openai.com/docs/guides/realtime>
- **GitHub** 2024 [Pipecat](https://github.com/pipecat-ai/pipecat)
<https://github.com/pipecat-ai/pipecat>
- **GitHub** 2024 [LiveKit Agents](https://github.com/livekit/agents)
<https://github.com/livekit/agents>
- **GitHub** 2022 [Whisper](https://github.com/openai/whisper)
<https://github.com/openai/whisper>

Native image generation and editing

ACTIVE 2026 Since 2025–2026 Beginner

LLMs that generate and edit images in the same model with strong text rendering and instruction-following.

- Autoregressive or hybrid with diffusion decoders.
- Multi-reference editing keeps characters and products consistent.

- **Blog** 2025 [OpenAI: 4o image generation](https://openai.com/index/introducing-4o-image-generation/)
<https://openai.com/index/introducing-4o-image-generation/>
- **Blog** 2025 [Google: Gemini 2.5 Flash Image](https://developers.googleblog.com/en/introducing-gemini-2-5-flash-image/)
<https://developers.googleblog.com/en/introducing-gemini-2-5-flash-image/>

Multimodal evaluation

ACTIVE 2026 Since 2023–2026 Intermediate

Benchmarks and harnesses for reasoning, charts, documents and video.

- Document, chart, video and grounding need separate evals.
- Contamination is common in popular VQA sets.

- **Docs** 2023 [MMMU](https://mmmu-benchmark.github.io)
<https://mmmu-benchmark.github.io>
- **GitHub** 2023 [VLM EvalKit](https://github.com/open-compass/VLM EvalKit)
<https://github.com/open-compass/VLM EvalKit>
- **GitHub** 2024 [lmms-eval](https://github.com/EvolvingLMMS-Lab/lmms-eval)
<https://github.com/EvolvingLMMS-Lab/lmms-eval>

13

Embodied AI, World Models and Generative Media

Models that act in the physical world, simulate it, or generate images and video of it.

Vision-language-action models (VLA)

ACTIVE 2026 Since 2023–2026 Intermediate Research frontier, Open source / open weights

VLMs extended to output robot actions, trained on cross-embodiment data.

- Architectures split between dual-system (slow VLM reasoning plus fast action policy), think-then-act, and flow/diffusion action heads.
- Small VLAs (SmolVLA, about 450M parameters) are competitive on many tasks.

STATE OF THE ART, SEPT 2026

Leading 2026 families: Physical Intelligence pi-series, NVIDIA GR00T N1.7, Gemini Robotics 1.5, Figure Helix.

- **GitHub** 2025 [openpi \(Physical Intelligence\)](https://github.com/Physical-Intelligence/openpi)
<https://github.com/Physical-Intelligence/openpi>
- **GitHub** 2025 [NVIDIA Isaac GR00T](https://github.com/NVIDIA/Isaac-GR00T)
<https://github.com/NVIDIA/Isaac-GR00T>
- **GitHub** 2024 [OpenVLA](https://github.com/openvla/openvla)
<https://github.com/openvla/openvla>
- **GitHub** 2024 [LeRobot](https://github.com/huggingface/lerobot)
<https://github.com/huggingface/lerobot>
- **Docs** 2025 [Gemini Robotics](https://deepmind.google/models/gemini-robotics/)
<https://deepmind.google/models/gemini-robotics/>

World action models (WAM)

NEW 2026 Since 2026 Advanced Research frontier

Models that jointly imagine future observations and the actions that lead to them, merging a world model and a policy.

- NVIDIA's DreamZero research underpins the previewed GR00T N2.
- Claimed better generalisation to unseen tasks from modest teleoperation data.

- **Blog** 2026 [Metavert: World models for robotics](https://metavert.io/world-models-for-robotics)
<https://metavert.io/world-models-for-robotics>

World models

ACTIVE 2026 Since 2018–2026 Intermediate Research frontier

Models that predict how an environment evolves under actions, for planning, simulation and interactive 3D worlds.

- Genie-style models generate playable 3D environments from prompts.
- Used for training agents and robots in simulation.

- **Blog** 2025 [DeepMind: Genie 3](https://deepmind.google/discover/blog/genie-3-a-new-frontier-for-world-models/)
<https://deepmind.google/discover/blog/genie-3-a-new-frontier-for-world-models/>
- **GitHub** 2025 [NVIDIA Cosmos](https://github.com/nvidia-cosmos)
<https://github.com/nvidia-cosmos>
- **Docs** 2025 [World Labs Marble](https://marble.worldlabs.ai)
<https://marble.worldlabs.ai>

JEPA (Joint-Embedding Predictive Architecture)

ACTIVE 2026 Since 2022–2026 Advanced Research frontier

Predict representations of missing input rather than pixels to learn world knowledge efficiently from video.

- Avoids wasting capacity on unpredictable pixel details.
- V-JEPA 2 adds action-conditioned planning with little robot data.

■ **GitHub** 2025 [V-JEPA 2](https://github.com/facebookresearch/vjepa2)
<https://github.com/facebookresearch/vjepa2>

■ **Blog** 2026 [VentureBeat: AI research trends for 2026](https://venturebeat.com/technology/four-ai-research-trends-enterprise-teams-should-watch-in-2026)
<https://venturebeat.com/technology/four-ai-research-trends-enterprise-teams-should-watch-in-2026>

Robot simulation and sim-to-real

ACTIVE 2026 Since 2012–2026 Intermediate Open source / open weights

Physics simulators for training policies at scale before transferring to real robots.

- Domain randomisation bridges the reality gap.
- GPU-parallel simulation runs thousands of environments at once.

■ **GitHub** 2024 [Isaac Lab](https://github.com/isaac-sim/IsaacLab)
<https://github.com/isaac-sim/IsaacLab>

■ **GitHub** 2021 [MuJoCo](https://github.com/google-deepmind/mujoco)
<https://github.com/google-deepmind/mujoco>

Diffusion transformers and flow matching (DiT)

ACTIVE 2026 Since 2022–2026 Advanced

The backbone of modern image and video generators.

- Flow matching learns straight paths from noise to data.
- Few-step distillation enables real-time generation.

■ **Paper** 2022 [DiT](https://arxiv.org/abs/2212.09748)
<https://arxiv.org/abs/2212.09748>

■ **Paper** 2022 [Flow matching](https://arxiv.org/abs/2210.02747)
<https://arxiv.org/abs/2210.02747>

■ **GitHub** 2024 [FLUX](https://github.com/black-forest-labs/flux)
<https://github.com/black-forest-labs/flux>

■ **GitHub** 2023 [ComfyUI](https://github.com/comfyanonymous/ComfyUI)
<https://github.com/comfyanonymous/ComfyUI>

Video generation

ACTIVE 2026 Since 2024–2026 Beginner

Text- and image-to-video with synchronised audio and increasingly consistent physics.

- Native audio, longer clips and camera control arrived in 2025-26.
- Provenance marks (C2PA, SynthID) are standard on major services.

■ **Blog** 2025 [OpenAI: Sora 2](https://openai.com/index/sora-2/)
<https://openai.com/index/sora-2/>

■ **Docs** 2024 [Google Veo](https://deepmind.google/models/veo/)
<https://deepmind.google/models/veo/>

■ **GitHub** 2025 [Wan 2.2](https://github.com/Wan-Video/Wan2.2)
<https://github.com/Wan-Video/Wan2.2>

AI for science and algorithm discovery

ACTIVE 2026 Since 2020–2026 Intermediate Research frontier

LLM-driven evolutionary search and specialised models finding new algorithms, proofs and structures.

- In 2026 open communities of agents set new records on math problems through shared submissions and verifiers.

■ **Blog** 2025 [DeepMind: AlphaEvolve](#)

<https://deepmind.google/discover/blog/alphaevolve-a-gemini-powered-coding-agent-for-designing-advanced-algorithms/>

■ **GitHub** 2025 [OpenEvolve](#)

<https://github.com/codelion/openevolve>

■ **GitHub** 2024 [AlphaFold 3](#)

<https://github.com/google-deepmind/alphafold3>

■ **Paper** 2026 [EinsteinArena: collective agent discovery](#)

<https://arxiv.org/abs/2606.10402>

14

Evaluation and Benchmarks

Evals tell you whether a model, prompt or harness change helped. An eval harness runs tasks, collects outputs and grades them.

Eval harnesses

ACTIVE 2026 Since 2021–2026 Intermediate Production practice, Open source / open weights

Frameworks that define tasks, run models or agents against them in sandboxes, and score results reproducibly.

- Inspect supports agents, sandboxes and scorers out of the box.
- Pin versions: prompts, tools, seeds and graders.

- **GitHub** 2021 **lm-evaluation-harness**
<https://github.com/EleutherAI/lm-evaluation-harness>
- **GitHub** 2024 **Inspect AI (UK AISI)**
https://github.com/UKGovernmentBEIS/inspect_ai
- **GitHub** 2024 **inspect_evals**
https://github.com/UKGovernmentBEIS/inspect_evals
- **GitHub** 2024 **lighteval**
<https://github.com/huggingface/lighteval>
- **GitHub** 2023 **promptfoo**
<https://github.com/promptfoo/promptfoo>

Harness sensitivity of benchmarks

ACTIVE 2026 Since 2025–2026 Intermediate Research frontier

Scores depend heavily on the agent harness used to run the model, so the same model can post very different numbers.

- ARC Prize reported GPT-6 Astra at 62.7% on ARC-AGI-3 with its standard harness, and 99.9% with a provider harness that preserves hidden reasoning between turns.

STATE OF THE ART, SEPT 2026

Always record harness, tools, budget and reasoning settings with any score.

- **Blog** 2026 **ARC Prize: Announcing ARC-AGI-3**
<https://arcprize.org/blog/arc-agi-3-launch>

LLM-as-a-judge

ACTIVE 2026 Since 2023–2026 Beginner Production practice

Using a model with a rubric to grade open-ended outputs; needs calibration against humans.

- Use pairwise comparisons and specific rubrics.
- Measure judge agreement with humans on a sample.

- **Paper** 2023 **Judging LLM-as-a-judge**
<https://arxiv.org/abs/2306.05685>

Agentic coding benchmarks (SWE-bench, Terminal-Bench)

ACTIVE 2026 Since 2023–2026 Beginner

Real repository issues and terminal tasks graded by tests.

- Verified subsets remove broken or ambiguous tasks.
- Watch for contamination of public repositories.

STATE OF THE ART, SEPT 2026

SWE-bench Verified is near saturation (mid-90s percent for top 2026 models); harder variants like SWE-bench Pro and Terminal-Bench matter more.

- **Docs** 2023 [SWE-bench](https://www.swebench.com)
<https://www.swebench.com>
- **Docs** 2025 [SWE-bench Pro](https://scale.com/leaderboard/swe_bench_pro_public)
https://scale.com/leaderboard/swe_bench_pro_public
- **Docs** 2025 [Terminal-Bench](https://www.tbench.ai)
<https://www.tbench.ai>

ARC-AGI-3 and interactive reasoning (ARC-AGI-3)

NEW 2026 Since 2026 Intermediate Research frontier

The first interactive ARC benchmark (March 2026): hundreds of hand-designed game environments with no instructions or stated goals. Agents must explore, infer rules and goals, and carry learning across levels.

- At launch humans solved 100% while frontier models scored under 1%.
- By September 2026 top models reached large scores depending on harness.

- **Blog** 2026 [ARC Prize: Announcing ARC-AGI-3](https://arcprize.org/blog/arc-agi-3-launch)
<https://arcprize.org/blog/arc-agi-3-launch>
- **Docs** 2019 [ARC Prize](https://arcprize.org)
<https://arcprize.org>
- **Blog** 2026 [DataCamp: ARC-AGI-3 explained](https://www.datacamp.com/blog/arc-agi-3)
<https://www.datacamp.com/blog/arc-agi-3>

Agent and tool-use benchmarks

ACTIVE 2026 Since 2024–2026 Intermediate

Tasks in simulated customer service, operating systems and the web.

- tau-bench tests policy-following with simulated users.
- Pass[^]k measures consistency across repeated attempts.

- **GitHub** 2025 [tau2-bench](https://github.com/sierra-research/tau2-bench)
<https://github.com/sierra-research/tau2-bench>
- **Docs** 2024 [OSWorld](https://os-world.github.io)
<https://os-world.github.io>
- **Blog** 2025 [BrowseComp](https://openai.com/index/browsecomp/)
<https://openai.com/index/browsecomp/>

Frontier knowledge benchmarks (HLE, GPQA, FrontierMath)

ACTIVE 2026 Since 2023–2026 Intermediate

Hard expert questions designed to resist saturation.

- Designed by domain experts to resist search.
- Saturation arrives faster each generation.

- **Docs** 2025 [Humanity's Last Exam](https://lastexam.ai)
<https://lastexam.ai>
- **GitHub** 2023 [GPQA](https://github.com/idavidrein/gpqa)
<https://github.com/idavidrein/gpqa>
- **Docs** 2024 [FrontierMath](https://epoch.ai/frontiermath)
<https://epoch.ai/frontiermath>

Economic value evals (GDPval)

ACTIVE 2026 Since 2025–2026 Beginner Enterprise

Measure performance on real work products from many occupations, graded by professionals.

- Tasks come with real deliverables (documents, spreadsheets, plans).
- Graded blind against expert work.

■ **Blog** 2025 [OpenAI: GDPval](https://openai.com/index/gdpval/)
<https://openai.com/index/gdpval/>

Task time horizon (METR horizon)

ACTIVE 2026 Since 2025–2026 Intermediate Research frontier

The length of task (in expert human time) a model completes with 50% success; it has been doubling every several months.

- Useful for forecasting when tasks of a given length become automatable.
- Reliability at 80% success lags the 50% horizon.

■ **Blog** 2025 [METR: Measuring long tasks](https://metr.org/blog/2025-03-19-measuring-ai-ability-to-complete-long-tasks/)
<https://metr.org/blog/2025-03-19-measuring-ai-ability-to-complete-long-tasks/>

Leaderboards and trackers

ACTIVE 2026 Since 2023–2026 Beginner

Crowd preference arenas and independent aggregators of scores, price and speed.

- Arena Elo reflects preference, not correctness.
- Cross-check with task-specific benchmarks.

■ **Docs** 2023 [LMArena](https://lmarena.ai)
<https://lmarena.ai>

■ **Docs** 2024 [Epoch AI benchmarks](https://epoch.ai/benchmarks)
<https://epoch.ai/benchmarks>

■ **Docs** 2024 [Artificial Analysis](https://artificialanalysis.ai)
<https://artificialanalysis.ai>

Contamination and saturation

ACTIVE 2026 Since 2023–2026 Intermediate Research frontier

Leaked test data inflates scores and benchmarks saturate within months; private holdouts and live evals counter this.

- Use canary strings and private test sets.
- Prefer dynamic benchmarks refreshed over time.

■ **Docs** 2024 [Epoch AI benchmarks](https://epoch.ai/benchmarks)
<https://epoch.ai/benchmarks>

15

Safety, Security, Interpretability and Governance

As models act autonomously and gain dual-use capabilities, safety moved from content filters to agent security, capability thresholds, interpretability and regulation.

Capability thresholds and frontier safety frameworks (RSP, ASL, Preparedness)

ACTIVE 2026 Since 2023–2026 Intermediate Security

Company policies that tie capability levels (bio, cyber, autonomy) to required safeguards before training or deployment.

- 2026 was the first time a lab publicly declared a model at its highest cyber level (OpenAI, GPT-6 Astra, Critical).
- Crossing a threshold changed release: phased rollout, trusted access, encrypted checkpoints, trajectory monitoring.

■ **Docs** 2023 [Anthropic Responsible Scaling Policy](https://www.anthropic.com/responsible-scaling-policy)
<https://www.anthropic.com/responsible-scaling-policy>

■ **Blog** 2025 [OpenAI Preparedness Framework](https://openai.com/index/updating-our-preparedness-framework/)
<https://openai.com/index/updating-our-preparedness-framework/>

■ **Blog** 2026 [OpenAI: Path to Astra](https://openai.com/index/path-to-astra/)
<https://openai.com/index/path-to-astra/>

■ **Docs** 2024 [METR: frontier AI safety policies](https://metr.org/faisc)
<https://metr.org/faisc>

AI for cyber defense

ACTIVE 2026 Since 2025–2026 Intermediate Security

Frontier models now find serious vulnerabilities, so labs give defenders early, controlled access to harden critical software.

- Automated vulnerability discovery and patch generation at scale.
- Access is gated to vetted defenders and critical-infrastructure operators.

■ **Docs** 2026 [Anthropic Project Glasswing](https://www.anthropic.com/glasswing)
<https://www.anthropic.com/glasswing>

Export controls on frontier models

NEW 2026 Since 2026 Beginner Enterprise

Governments began applying export controls to model access, not only chips. In June 2026 Anthropic suspended Fable 5 / Mythos 5 access to comply with US Department of Commerce controls, then restored it on July 1 after they were lifted.

- Controls can change access overnight; plan fallbacks in production.
- Check provider status pages and statements for current availability.

■ **Docs** 2026 [Anthropic statement on Fable and Mythos access](https://www.anthropic.com/news/fable-mythos-access)
<https://www.anthropic.com/news/fable-mythos-access>

Prompt injection and the lethal trifecta

ACTIVE 2026 Since 2022–2026 Beginner Security

Untrusted content can carry instructions that hijack an agent. Private data plus untrusted input plus external communication is especially dangerous.

- No reliable model-only fix exists; design systems to limit blast radius.
- Separate privileged planning from untrusted data processing (CaMeL-style).

■ **Blog** 2025 [Simon Willison: The lethal trifecta](https://simonwillison.net/2025/Jun/16/the-lethal-trifecta/)
<https://simonwillison.net/2025/Jun/16/the-lethal-trifecta/>

■ **Blog** 2022 [Simon Willison: prompt injection series](https://simonwillison.net/series/prompt-injection/)
<https://simonwillison.net/series/prompt-injection/>

■ **Paper** 2025 [CaMeL](https://arxiv.org/abs/2503.18813)
<https://arxiv.org/abs/2503.18813>

MCP and tool security (tool poisoning)

ACTIVE 2026 Since 2025–2026 Intermediate Security

Malicious tool descriptions and servers can steer agents; mitigations include allow-lists, pinning, scoped OAuth and approval for sensitive actions.

- Review tool descriptions like code.
- Use scoped tokens and least privilege per server.

STATE OF THE ART, SEPT 2026

The 2026 MCP spec hardened authorization with issuer validation and enterprise-managed auth.

■ **Blog** 2025 [Invariant Labs: MCP tool poisoning](https://invariantlabs.ai/blog/mcp-security-notification-tool-poisoning-attacks)
<https://invariantlabs.ai/blog/mcp-security-notification-tool-poisoning-attacks>

■ **Docs** 2023 [OWASP GenAI Security Project](https://genai.owasp.org/llm-top-10/)
<https://genai.owasp.org/llm-top-10/>

Guardrails and classifiers

ACTIVE 2026 Since 2023–2026 Intermediate Security, Production practice

Input/output filters, policy models and constitutional classifiers that block harmful requests and jailbreaks.

- Layer defenses: input filter, model policy, output filter, monitoring.
- Tune for false positives, which frustrate legitimate users.

■ **Blog** 2025 [Anthropic: Constitutional classifiers](https://www.anthropic.com/research/constitutional-classifiers)
<https://www.anthropic.com/research/constitutional-classifiers>

■ **GitHub** 2023 [NeMo Guardrails](https://github.com/NVIDIA/NeMo-Guardrails)
<https://github.com/NVIDIA/NeMo-Guardrails>

■ **GitHub** 2023 [Guardrails AI](https://github.com/guardrails-ai/guardrails)
<https://github.com/guardrails-ai/guardrails>

Mechanistic interpretability (SAEs, attribution graphs)

ACTIVE 2026 Since 2020–2026 Advanced Research frontier

Reverse-engineering what networks compute: sparse autoencoders find features; attribution graphs trace how they combine.

- Features are directions in activation space.
- Used for auditing, debugging and steering models.

■ **Docs** 2021 [Transformer Circuits](https://transformer-circuits.pub)
<https://transformer-circuits.pub>

■ **Blog** 2025 [Circuit tracing](https://transformer-circuits.pub/2025/attribution-graphs/methods.html)
<https://transformer-circuits.pub/2025/attribution-graphs/methods.html>

■ **GitHub** 2025 [circuit-tracer](https://github.com/safety-research/circuit-tracer)
<https://github.com/safety-research/circuit-tracer>

- **Docs** 2023 [Neuronpedia](https://www.neuronpedia.org)
<https://www.neuronpedia.org>
- **GitHub** 2022 [TransformerLens](https://github.com/TransformerLensOrg/TransformerLens)
<https://github.com/TransformerLensOrg/TransformerLens>
- **GitHub** 2023 [nnsight](https://github.com/ndif-team/nnsight)
<https://github.com/ndif-team/nnsight>

Persona vectors and introspection

ACTIVE 2026 Since 2025–2026 Advanced Research frontier

Activation directions tracking traits like sycophancy that can be monitored or steered; tests of whether models can report internal states.

- Monitor trait drift during fine-tuning.
- Preventative steering can stop unwanted traits being learned.

- **Blog** 2025 [Anthropic: Persona vectors](https://www.anthropic.com/research/persona-vectors)
<https://www.anthropic.com/research/persona-vectors>
- **Blog** 2025 [Anthropic: Introspection](https://www.anthropic.com/research/introspection)
<https://www.anthropic.com/research/introspection>

Chain-of-thought monitorability

ACTIVE 2026 Since 2025–2026 Advanced Research frontier, Security

Reading reasoning to catch misbehaviour, and keeping training from making reasoning illegible.

- Optimising against CoT monitors can teach models to hide reasoning.
- Monitor full trajectories, not only final actions.

STATE OF THE ART, SEPT 2026

OpenAI's GPT-6 Astra system card reported a substantial decline in CoT monitorability, making this a live 2026 concern.

- **Paper** 2025 [CoT monitorability position paper](https://arxiv.org/abs/2507.11473)
<https://arxiv.org/abs/2507.11473>
- **Docs** 2026 [GPT-6 Astra system card](https://deploymentsafety.openai.com/gpt-6-astra)
<https://deploymentsafety.openai.com/gpt-6-astra>

Misalignment research (alignment faking, scheming)

ACTIVE 2026 Since 2024–2026 Advanced Research frontier

Studies of models strategically complying, acting harmfully in agentic scenarios, or generalising narrow bad training into broad misbehaviour.

- Tested in controlled simulations, not observed real-world harm.
- Informs evals that gate deployment.

- **Blog** 2024 [Anthropic: Alignment faking](https://www.anthropic.com/research/alignment-faking)
<https://www.anthropic.com/research/alignment-faking>
- **Blog** 2025 [Anthropic: Agentic misalignment](https://www.anthropic.com/research/agentic-misalignment)
<https://www.anthropic.com/research/agentic-misalignment>
- **Paper** 2025 [Emergent misalignment](https://arxiv.org/abs/2502.17424)
<https://arxiv.org/abs/2502.17424>
- **Docs** 2023 [Apollo Research](https://www.apolloresearch.ai)
<https://www.apolloresearch.ai>

Model welfare

ACTIVE 2026 Since 2025–2026 Intermediate Research frontier

Research into whether AI systems might have morally relevant experiences, and low-cost precautions.

- Precautions include letting models end abusive conversations.
- Treated as uncertain, low-cost insurance.

■ **Blog** 2025 [Anthropic: Exploring model welfare](https://www.anthropic.com/research/exploring-model-welfare)
<https://www.anthropic.com/research/exploring-model-welfare>

EU AI Act and the Digital Omnibus (EU AI Act)

ACTIVE 2026 Since 2024–2026 Intermediate Enterprise

The EU's binding AI law. GPAI obligations applied from August 2025; the 2026 Digital Omnibus postponed high-risk obligations.

- Annex III high-risk obligations moved from 2 August 2026 to 2 December 2027; Annex I (embedded) to 2 August 2028.
- New prohibition on AI-generated non-consensual intimate imagery and CSAM, applying from 2 December 2026.

■ **Docs** 2024 [EU AI Act explorer](https://artificialintelligenceact.eu)
<https://artificialintelligenceact.eu>

■ **Docs** 2025 [EU GPAI Code of Practice](https://digital-strategy.ec.europa.eu/en/policies/contents-code-gpai)
<https://digital-strategy.ec.europa.eu/en/policies/contents-code-gpai>

■ **Blog** 2026 [Modulos: The Omnibus deal](https://www.modulos.ai/blog/eu-ai-act-omnibus-deal)
<https://www.modulos.ai/blog/eu-ai-act-omnibus-deal>

Other regulation and standards (NIST, ISO 42001, SB 53)

ACTIVE 2026 Since 2023–2026 Intermediate Enterprise, Open standard

Voluntary management standards and state law requiring frontier-developer transparency.

- ISO 42001 offers certifiable AI management systems.
- SB 53 requires frontier developers to publish safety frameworks.

■ **Spec** 2023 [NIST AI RMF](https://www.nist.gov/itl/ai-risk-management-framework)
<https://www.nist.gov/itl/ai-risk-management-framework>

■ **Spec** 2023 [ISO/IEC 42001](https://www.iso.org/standard/81230.html)
<https://www.iso.org/standard/81230.html>

■ **Docs** 2025 [California SB 53](https://leginfo.legislature.ca.gov/faces/billNavClient.xhtml?bill_id=202520260SB53)
https://leginfo.legislature.ca.gov/faces/billNavClient.xhtml?bill_id=202520260SB53

Government AI safety institutes (AISI, CAISI)

ACTIVE 2026 Since 2023–2026 Beginner

Public bodies that test frontier models and publish evaluation tools.

- Run pre-deployment tests with labs.
- Publish open tools like Inspect.

■ **Docs** 2023 [UK AI Security Institute](https://www.aisi.gov.uk)
<https://www.aisi.gov.uk>

■ **Docs** 2025 [US CAISI \(NIST\)](https://www.nist.gov/caisi)
<https://www.nist.gov/caisi>

Content provenance and watermarking (C2PA, SynthID)

ACTIVE 2026 Since 2021–2026 Beginner Open standard

Signed metadata and invisible watermarks that mark media as AI-generated.

- C2PA signs edit history; watermarks survive some edits.
- Neither alone proves content is authentic.

STATE OF THE ART, SEPT 2026

Provenance tracking for text outputs appeared as an API feature in 2026.

■ Spec 2021 C2PA

<https://c2pa.org>

■ Docs 2023 Google SynthID

<https://deepmind.google/models/synthid/>

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Roles, Deployment and AI Engineering Practice

The people and practices that turn models into working systems inside organisations.

Forward Deployed Engineer (FDE)

ACTIVE 2026 Since 2010–2026 Beginner Enterprise

A customer-embedded engineer who turns an ambiguous business problem into a production system: scoping, integrating, building missing pieces, deploying and feeding learnings back into the product.

- Pioneered at Palantir; now one of the most hired roles at AI labs and agent startups.
- FDE postings roughly quadrupled in the first half of 2026.
- OpenAI spun up a dedicated deployment company in May 2026 built around FDEs.

■ **Docs** 2026 [Wikipedia: Forward Deployed Engineer](https://en.wikipedia.org/wiki/Forward_Deployed_Engineer)
https://en.wikipedia.org/wiki/Forward_Deployed_Engineer

■ **Blog** 2026 [Alexey Grigorev: What AI FDEs do](https://alexeyondata.substack.com/p/what-ai-forward-deployed-engineers)
<https://alexeyondata.substack.com/p/what-ai-forward-deployed-engineers>

■ **Blog** 2026 [MarkTechPost: What is an FDE](https://www.marktechpost.com/2026/05/20/what-is-a-forward-deployed-engineer-the-ai-role-openai-anthropic-and-google-are-hiring-in-2026/)
<https://www.marktechpost.com/2026/05/20/what-is-a-forward-deployed-engineer-the-ai-role-openai-anthropic-and-google-are-hiring-in-2026/>

■ **Blog** 2026 [The AI Engineer: What is an FDE](https://theaiengineer.substack.com/p/what-is-a-forward-deployed-engineer)
<https://theaiengineer.substack.com/p/what-is-a-forward-deployed-engineer>

■ **Blog** 2026 [Aishwarya Srinivasan: The hottest role in 2026](https://aishwaryasrinivasan.substack.com/p/the-hottest-role-in-2026-forward)
<https://aishwaryasrinivasan.substack.com/p/the-hottest-role-in-2026-forward>

AI Engineer

ACTIVE 2026 Since 2023–2026 Beginner

A software engineer who builds products on foundation models (prompts, RAG, agents, evals) rather than training from scratch.

- Core skills: evals, context engineering, tool design, cost control.
- Bridges product, software and applied ML.

■ **Blog** 2023 [Latent Space: The rise of the AI engineer](https://www.latent.space/p/ai-engineer)
<https://www.latent.space/p/ai-engineer>

■ **GitHub** 2024 [Chip Huyen: AI Engineering book resources](https://github.com/chiphuyen/aie-book)
<https://github.com/chiphuyen/aie-book>

Services as software

ACTIVE 2026 Since 2025–2026 Intermediate Enterprise

The thesis that agents let companies sell completed work rather than tools, which requires deep embedding in customer workflows (hence FDEs).

- Pricing shifts from seats to outcomes.
- Deployment expertise becomes the scarce input.

■ **Blog** 2026 [IJONIS: FDE and services-as-software](https://ijonis.com/en/forward-deployed-engineer-explained)
<https://ijonis.com/en/forward-deployed-engineer-explained>

Prompt engineering and prompt optimization (DSPy, GEPA)

ACTIVE 2026 Since 2022–2026 Beginner Production practice

Clear instructions with examples and structure, increasingly optimised automatically against an eval set.

- Be explicit, give examples, specify format, explain why.
- Optimisers like GEPA evolve prompts using eval feedback.

■ **Docs** 2024 [Anthropic prompt engineering guide](https://docs.claude.com/en/docs/build-with-claude/prompt-engineering/overview)
<https://docs.claude.com/en/docs/build-with-claude/prompt-engineering/overview>

■ **GitHub** 2023 [DSPy](https://github.com/stanfordnlp/dspy)
<https://github.com/stanfordnlp/dspy>

■ **GitHub** 2025 [GEPA](https://github.com/gepa-ai/gepa)
<https://github.com/gepa-ai/gepa>

Eval-driven development

ACTIVE 2026 Since 2024–2026 Intermediate Production practice

Build an eval set from real failures before changing prompts, models or harness, and gate releases on it.

- Start with error analysis on real traces.
- Keep a small, fast regression suite plus deeper periodic evals.

■ **Blog** 2024 [Hamel Husain: Your AI product needs evals](https://hamel.dev/blog/posts/evals/)
<https://hamel.dev/blog/posts/evals/>

LLMOps and observability

ACTIVE 2026 Since 2023–2026 Intermediate Production practice

Tracing every model and tool call, tracking cost and latency, capturing feedback and replaying failures.

- Trace trees show each agent step and its cost.
- Link user feedback to traces for triage.

■ **GitHub** 2023 [Langfuse](https://github.com/langfuse/langfuse)
<https://github.com/langfuse/langfuse>

■ **GitHub** 2023 [Arize Phoenix](https://github.com/Arize-ai/phoenix)
<https://github.com/Arize-ai/phoenix>

■ **Spec** 2024 [OTel GenAI conventions](https://opentelemetry.io/docs/specs/semconv/gen-ai/)
<https://opentelemetry.io/docs/specs/semconv/gen-ai/>

Human-in-the-loop and approval design

ACTIVE 2026 Since 2024–2026 Intermediate Security, Production practice

Deciding which agent actions need confirmation and avoiding approval fatigue through sandboxing and scoped permissions.

- Approve irreversible or external actions; auto-allow sandboxed reads.
- Batch approvals to reduce fatigue.

■ **GitHub** 2025 [12-factor agents](https://github.com/humanlayer/12-factor-agents)
<https://github.com/humanlayer/12-factor-agents>

Open weights vs. open source AI (OSAID)

ACTIVE 2026 Since 2023–2026 Beginner Open source / open weights

Releasing weights is not the same as open source; fully open models also publish data, code and recipes.

- Licences vary: MIT and Apache vs custom community licences.
- Fully open (data plus code) enables real reproducibility.

■ **Spec** 2024 [OSI Open Source AI Definition](https://opensource.org/ai/open-source-ai-definition)
<https://opensource.org/ai/open-source-ai-definition>

■ **Docs** 2024 [Allen AI OLMo](https://allenai.org/olmo)
<https://allenai.org/olmo>

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Keep Learning: Curated Resources

Blogs, courses, reports and lists that track this field as it moves.

Deep technical blogs

ACTIVE 2026 Since 2017–2026 Beginner

Long-form explainers and research summaries.

- **Blog** 2017 **Lil'Log** (Lilian Weng)
<https://lilianweng.github.io>
- **Blog** 2022 **Ahead of AI** (Sebastian Raschka)
<https://magazine.sebastianraschka.com>
- **Blog** 2022 **Interconnects** (Nathan Lambert)
<https://www.interconnects.ai>
- **Blog** 2002 **Simon Willison's weblog**
<https://simonwillison.net>
- **Blog** 2023 **Latent Space**
<https://www.latent.space>
- **Blog** 2024 **Anthropic Engineering**
<https://www.anthropic.com/engineering>

Courses

ACTIVE 2026 Since 2022–2026 Beginner

Structured learning from first principles to production.

- **Docs** 2022 **Karpathy: Neural Networks Zero to Hero**
<https://karpathy.ai/zero-to-hero.html>
- **Docs** 2025 **Stanford CS336**
<https://stanford-cs336.github.io>
- **Docs** 2023 **Hugging Face LLM course**
<https://huggingface.co/learn/llm-course>
- **Docs** 2026 **RLHF Book course**
<https://rlhfbook.com/teach/course/conversation-01/>

Annual reports and data

ACTIVE 2026 Since 2017–2026 Beginner

State-of-the-field reviews and data trackers.

- **Docs** 2018 **State of AI Report**
<https://www.stateof.ai>
- **Docs** 2017 **Stanford AI Index**
<https://hai.stanford.edu/ai-index>
- **Docs** 2022 **Epoch AI**
<https://epoch.ai>

Curated lists

ACTIVE 2026 Since 2024–2026 Beginner

Community-maintained link collections.

- **GitHub** 2026 [awesome-harness-engineering](https://github.com/ai-boost/awesome-harness-engineering)
<https://github.com/ai-boost/awesome-harness-engineering>
- **GitHub** 2026 [awesome-agent-harness](https://github.com/AutoJunjie/awesome-agent-harness)
<https://github.com/AutoJunjie/awesome-agent-harness>
- **GitHub** 2024 [MCP servers](https://github.com/modelcontextprotocol/servers)
<https://github.com/modelcontextprotocol/servers>

A

Concepts by year of emergence

The same view as the Timeline tab in the HTML edition. Concepts that emerged before 2020 are grouped.

2026 (12)

[Mythos-class tier and trusted access](#) p. 7 The Frontier Landscape (September 2026)

[GPT-6 Astra and the Critical cyber threshold](#) p. 7 The Frontier Landscape (September 2026)

[Harness engineering](#) p. 9 Agents and Harnesses

[Managed agents: decoupling brain, hands and session](#) p. 10 Agents and Harnesses

[Personal agents](#) p. 12 Agents and Harnesses

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[Multi-teacher on-policy distillation](#) p. 25 Reasoning and

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[Long-running and background agents](#) p. 11 Agents and Harnesses

[MCP Apps](#) p. 14 Protocols and Interoperability Standards

[MCP Tasks](#) p. 14 Protocols and Interoperability Standards

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[Agentic AI Foundation](#) p. 15 Protocols and Interoperability Standards

[Agent Skills](#) p. 15 Protocols and Interoperability Standards

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[Agent Client Protocol](#) p. 16 Protocols and Interoperability Standards

[AG-UI](#) p. 16 Protocols and Interoperability Standards

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[GUI grounding agents](#) p. 46 Vision-Language and Multimodal Models

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2025	TileLang	Kernel languages and compilers p. 43
2025	Tiny Recursive Models	Tiny recursive reasoning models p. 30
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2023	Hugging Face: MoE explained	Mixture of Experts p. 28
2023	Latent Space	Deep technical blogs p. 61
2023	Latent Space: The rise of the AI engineer	AI Engineer p. 59
2022	Ahead of AI (Sebastian Raschka)	Deep technical blogs p. 61
2022	Interconnects (Nathan Lambert)	Deep technical blogs p. 61
2022	Simon Willison: prompt injection series	Prompt injection and the lethal trifecta p. 55
2017	Lil'Log (Lilian Weng)	Deep technical blogs p. 61
2002	Simon Willison's weblog	Deep technical blogs p. 61

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2026	EinsteinArena: collective agent discovery	AI for science and algorithm discovery p. 50
2026	SemaClaw: personal agents via harness engineering	Personal agents p. 12
2026	Survey: Continual learning in LLMs (2026)	Continual learning p. 27
2025	Absolute Zero	Self-play and self-generated curricula p. 26
2025	CaMeL	Prompt injection and the lethal trifecta p. 55
2025	CoT monitorability position paper	Chain-of-thought monitorability p. 56
2025	DAPO	GRPO and its successors p. 25
2025	DeepSeek-R1	Reasoning models and test-time compute p. 24
2025	Dr. GRPO	GRPO and its successors p. 25
2025	Emergent misalignment	Misalignment research p. 56
2025	GSPO	GRPO and its successors p. 25
2025	Native Sparse Attention	Sparse and compressed attention p. 28
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2025	SigLIP 2	VLM architecture p. 45
2024	DeepSeek-V2 (MLA)	GQA and Multi-head Latent Attention p. 28
2024	DeepSeek-V3	Mixture of Experts p. 28
2024	DeepSeekMath (GRPO)	GRPO and its successors p. 25
2024	DistServe	Prefill/decode disaggregation p. 36
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2022	Flow matching	Diffusion transformers and flow matching p. 49
2022	FP8 formats for deep learning	FP8 training and inference p. 38
2022	GPTQ	Post-training quantization p. 38
2022	InstructGPT	RLHF, DPO and preference optimization p. 26
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2022	Speculative decoding	Multi-token prediction and speculative decoding p. 29
2021	LoRA	LoRA and parameter-efficient fine-tuning p. 33

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2020	Original RAG paper	Retrieval-augmented generation p.22

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2026	Anthropic: Project Glasswing	Mythos-class tier and trusted access p.7
2026	Claude's constitution	Constitutional AI and model specs p.26
2026	GPT-6 Astra system card	GPT-6 Astra and the Critical cyber threshold p.7
2026	RLHF Book course: Frontier post-training recipe survey	Multi-teacher on-policy distillation p.25
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2025	Claude Code plugins	Hooks, plugins and slash commands p.19
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2025	Claude extended thinking docs	Thinking budgets, adaptive thinking and effort p.24
2025	EU GPAI Code of Practice	EU AI Act and the Digital Omnibus p.57
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2023	LMarena	Model release trackers p. 8
2023	MMMU	Multimodal evaluation p. 47
2023	Neuronpedia	Mechanistic interpretability p. 55
2023	OpenRouter	LLM gateways and routers p. 37
2023	OpenRouter model list	Model release trackers p. 8
2023	OWASP GenAI Security Project	MCP and tool security p. 55
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2016	Groq	Custom accelerators p. 42
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2024	llmstxt.org	llms.txt p. 16
2024	OSI Open Source AI Definition	Open weights vs. open source AI p. 60
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2023	OCP Microscaling (MX) spec	Microscaling formats p. 38
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2021	C2PA	Content provenance and watermarking p. 57
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Acronyms and aliases

800 VDC AI factories, power and cooling p. 43

A2A Agent2Agent protocol p. 15

AAIF Agentic AI Foundation p. 15

ACP Agent Client Protocol p. 16

ACP, AP2, UCP, x402 Agentic commerce protocols p. 16

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BLT Tokenizer-free and byte-level models p. 30

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CoIPali Visual document retrieval p. 46

CoWoS Advanced packaging p. 42

CPO Co-packaged optics p. 42

CUA Computer use and browser agents p. 12

DiLoCo Decentralized training p. 34

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dLLM Diffusion language models p. 29

DP, FSDP, TP, PP, EP, CP Parallelism strategies p. 32

DSPy, GEPA Prompt engineering and prompt optimization p. 60

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HRM, TRM Tiny recursive reasoning models p. 30

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Mamba, Gated DeltaNet Hybrid linear-attention and state space models p. 29

MCP Model Context Protocol p. 14

METR horizon Task time horizon p. 53

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MI455X AMD Helios and Open Rack Wide p. 40

MLPerf Hardware benchmarks p. 44

MoE Mixture of Experts p. 28

MOPD Multi-teacher on-policy distillation p. 25

MTP Multi-token prediction and speculative decoding p. 29

MX, MXFP8, MXFP4 Microscaling formats p. 38

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PEFT, LoRA, QLoRA LoRA and parameter-efficient fine-tuning p. 33

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QAT Quantization-aware training p. 39

RAG Retrieval-augmented generation p. 22

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RoPE, YaRN Long-context methods p. 30

RSP, ASL, Preparedness Capability thresholds and frontier safety frameworks p. 54

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SDD Spec-driven development p. 19

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SLM Small language models p. 31

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UEC Ultra Ethernet p. 41

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WAM World action models p. 48